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Magno et al.

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(45) **Date of Patent: Aug. 26, 2025**

(54) **ARTICULATING DEBRIDER BLADE TIP AND HANDPIECE CONTROL**

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(73) Assignee: **Gyrus ACMI, Inc.**, Westborough, MA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 377 days.

(21) Appl. No.: **17/814,979**

(22) Filed: **Jul. 26, 2022**

(65) **Prior Publication Data**

US 2023/0030109 A1 Feb. 2, 2023

Related U.S. Application Data

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(51) **Int. Cl.**
A61B 17/32 (2006.01)

(52) **U.S. Cl.**
CPC *A61B 17/32002* (2013.01)

(58) **Field of Classification Search**
CPC A61B 17/0469; A61B 17/00234; A61B 17/0487; A61B 17/068; A61B 17/2909; A61B 17/320016; A61B 17/32002; A61B 2017/2908; A61B 2017/2929; A61B 2017/0647; A61B 2017/00309; A61B 2017/0488
See application file for complete search history.

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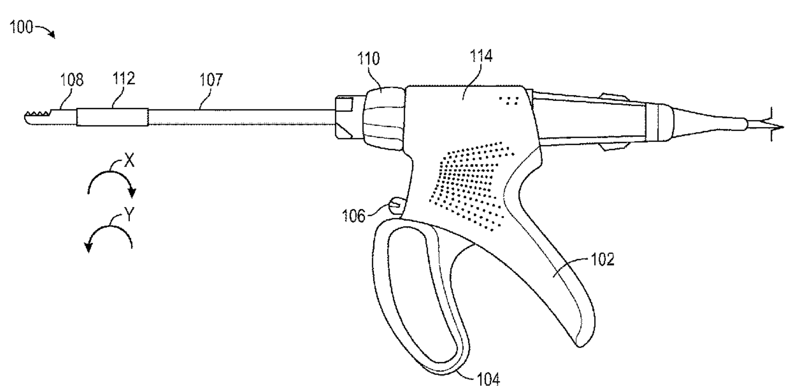
Primary Examiner — Tuan V Nguyen

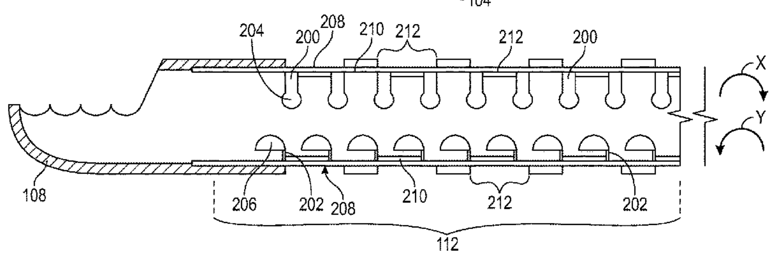
(74) *Attorney, Agent, or Firm* — Schwegman Lundberg & Woessner, P.A.

(57) **ABSTRACT**

A handheld surgical instrument having a handpiece and a shaft extending from the handpiece is provided. The instrument includes a cutting implement disposed at a distal end of the shaft and an articulation portion disposed proximate to the cutting implement. The articulation portion includes a set of first slits and a set of second slits opposite the first set of slits. The articulation portion also has first and second passageways disposed in the slits along with first and second flat pull wires disposed in the passageways. The instrument also has an articulation control assembly that includes an articulator having first and second anchor points coupled with the first and second flat pull wires. The articulator moves the first flat pull wire in a first direction and moves the second flat pull wire in a second direction opposite the first direction during actuation.

11 Claims, 29 Drawing Sheets





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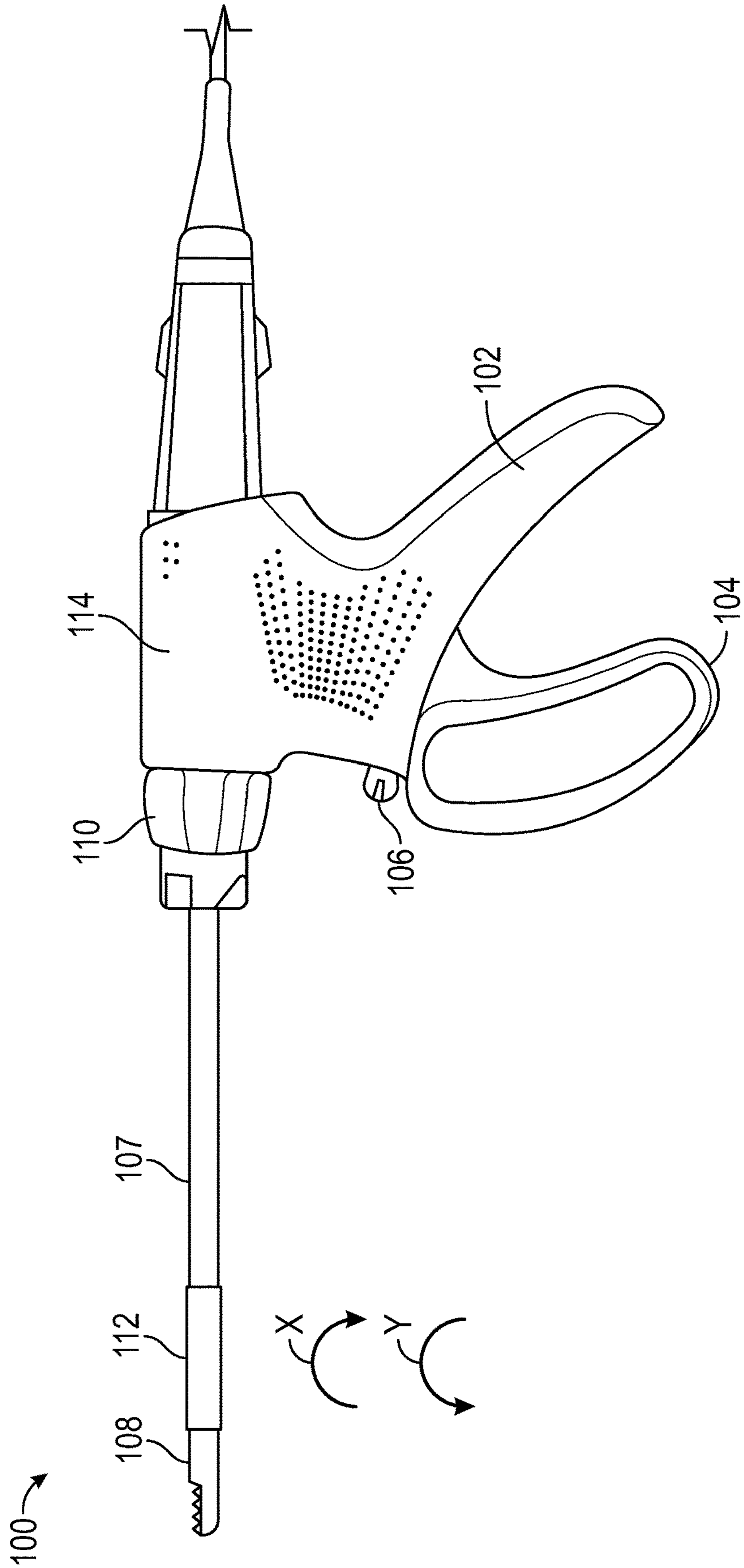


FIG. 1

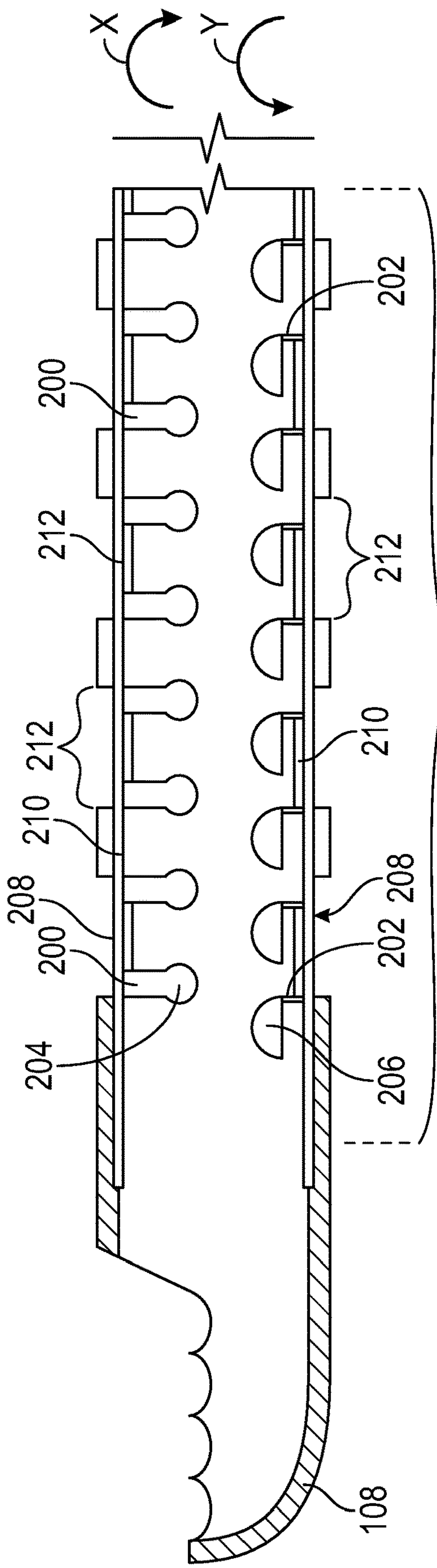


FIG. 2

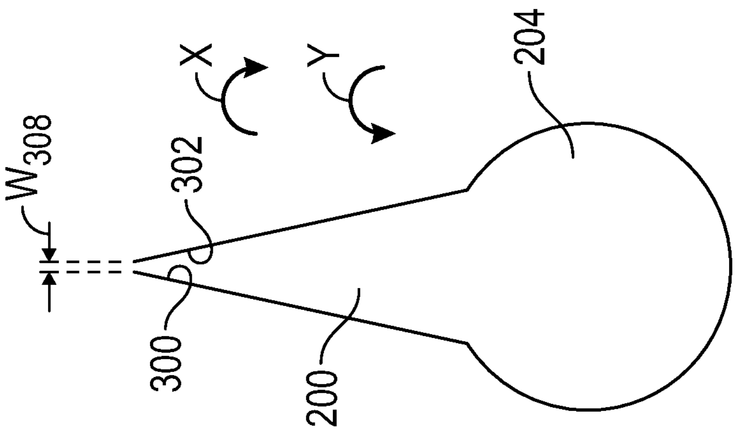


FIG. 3C

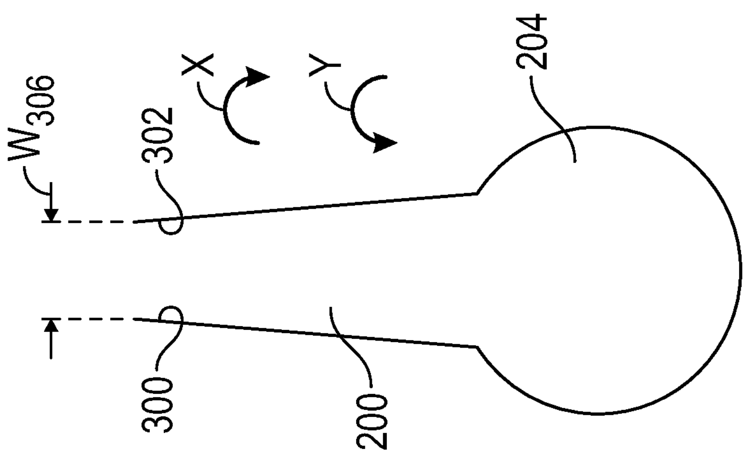


FIG. 3B

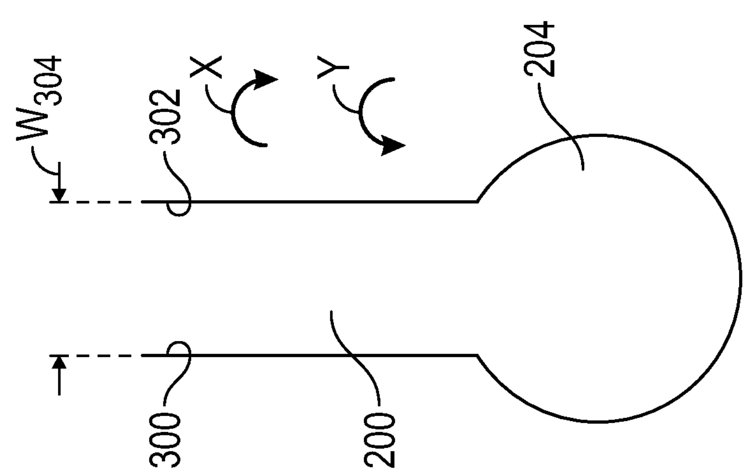


FIG. 3A

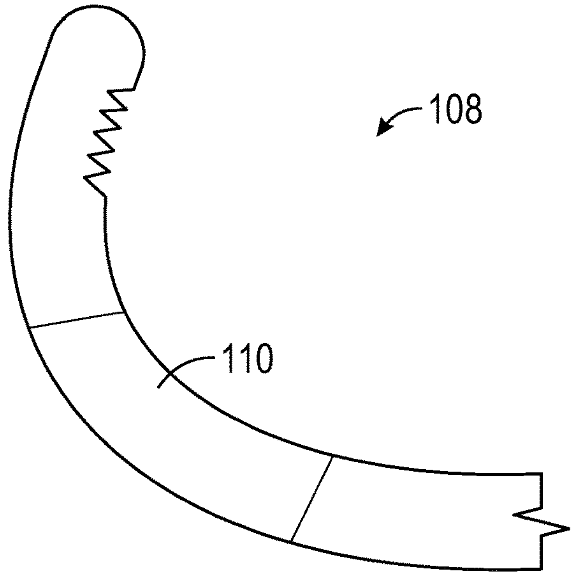


FIG. 4A

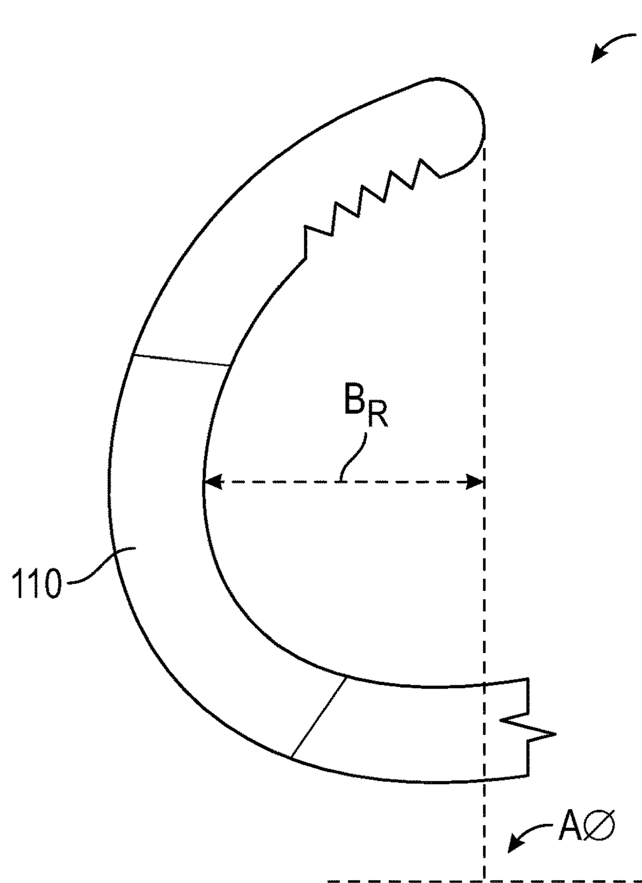


FIG. 4B

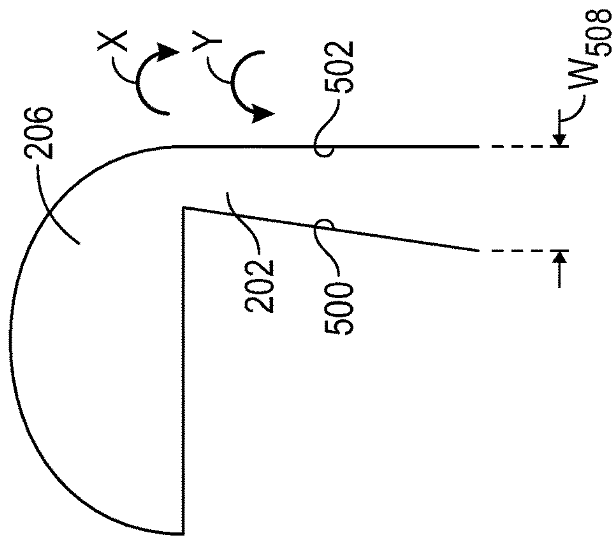


FIG. 5C

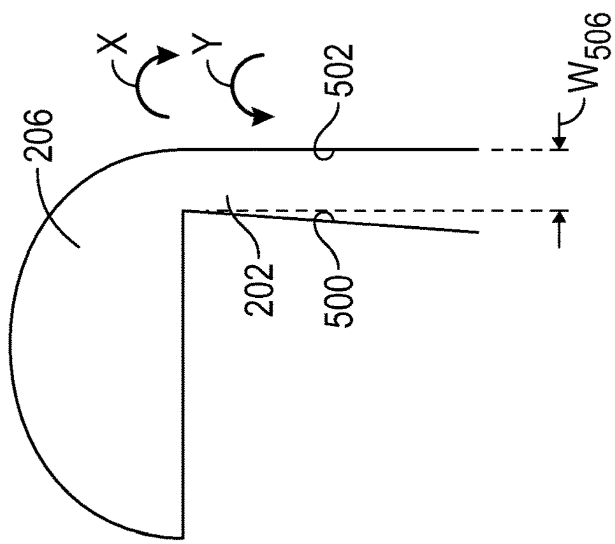


FIG. 5B

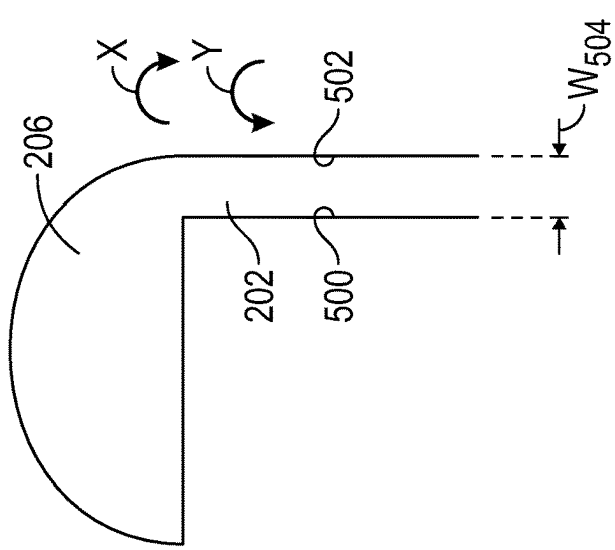
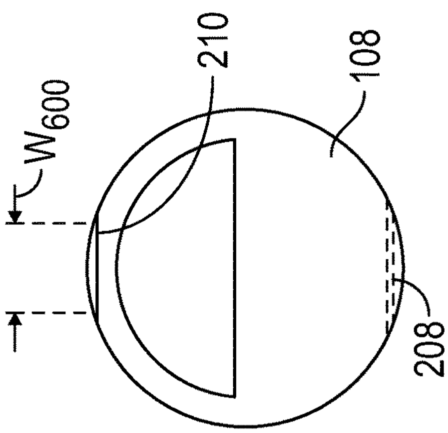


FIG. 5A



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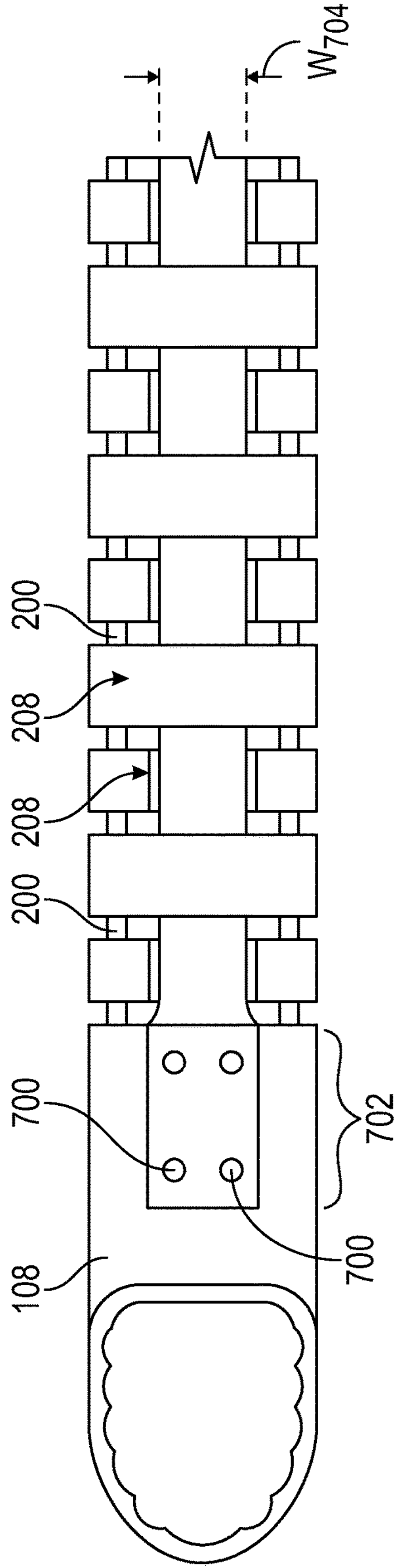

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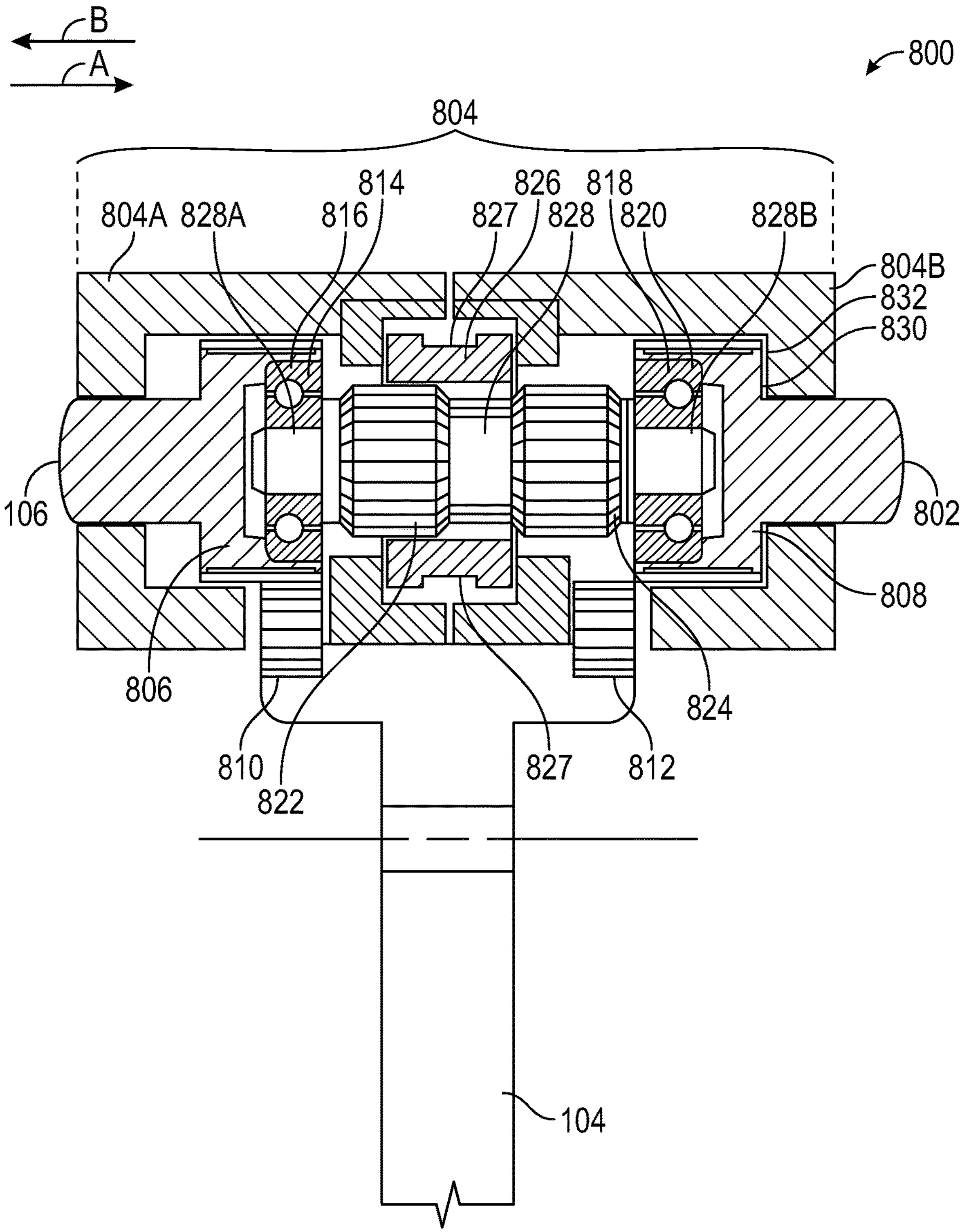
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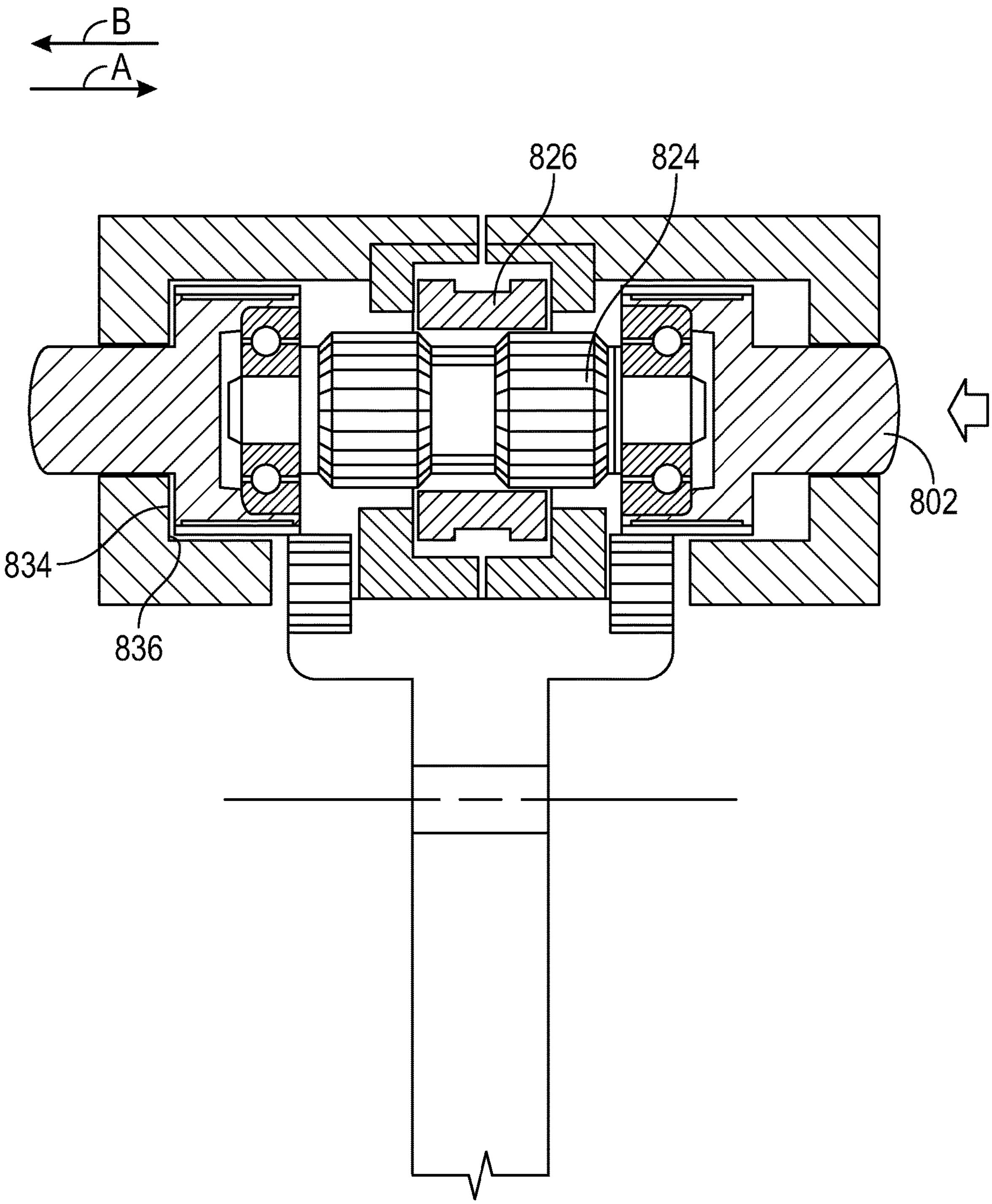
FIG. 8A



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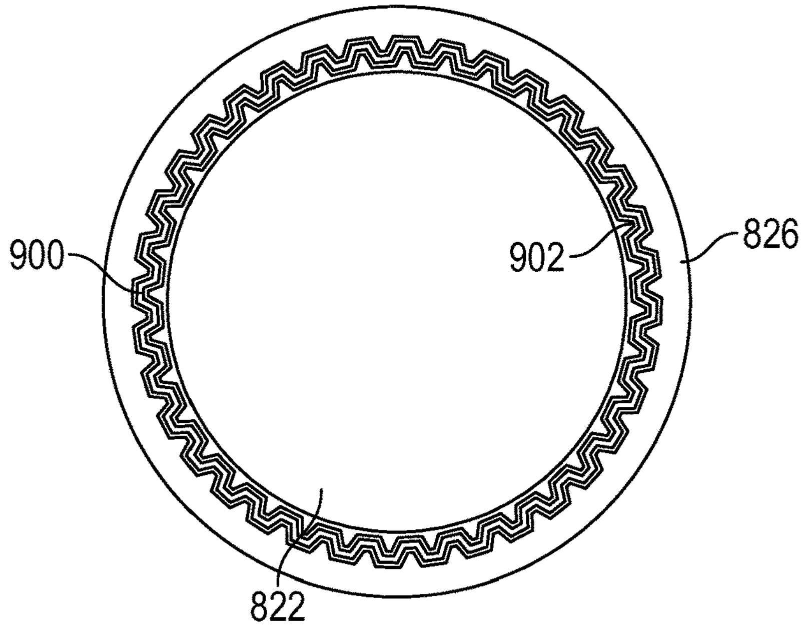
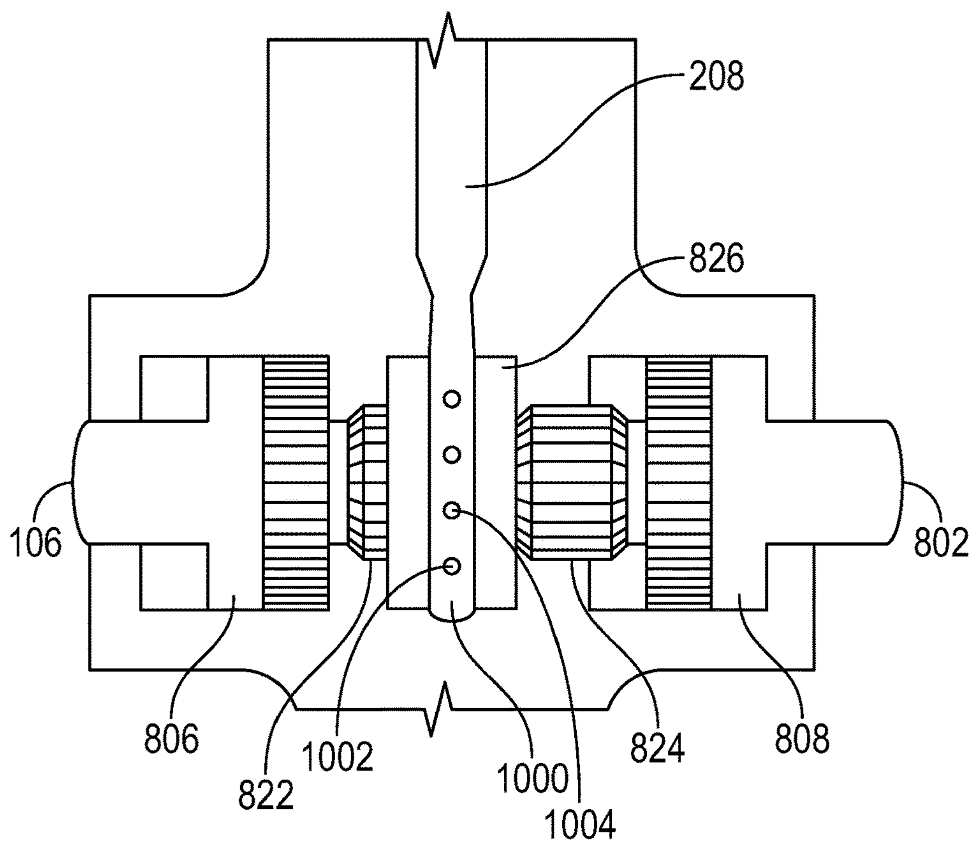


FIG. 9



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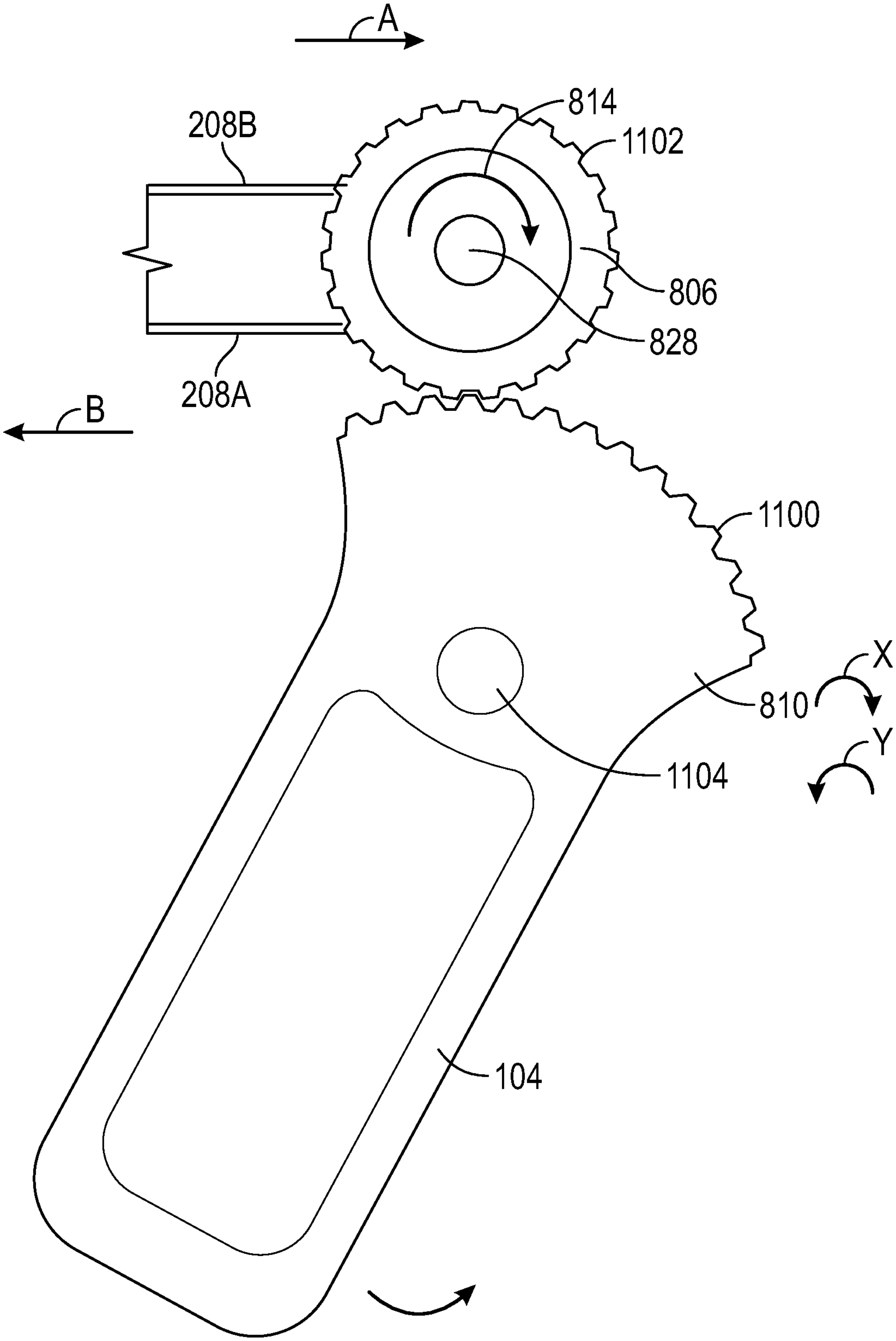

FIG. 11A

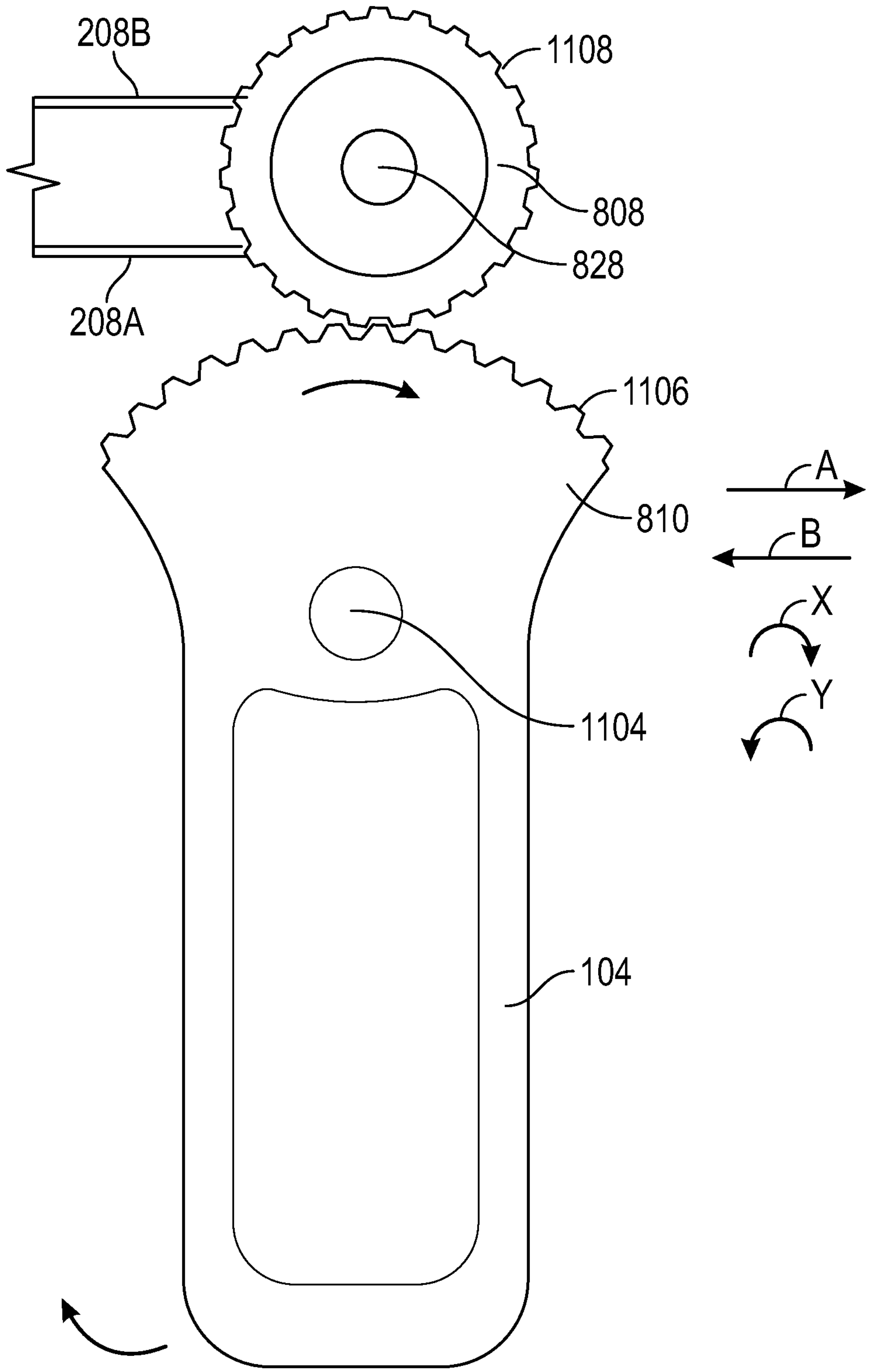

FIG. 11B

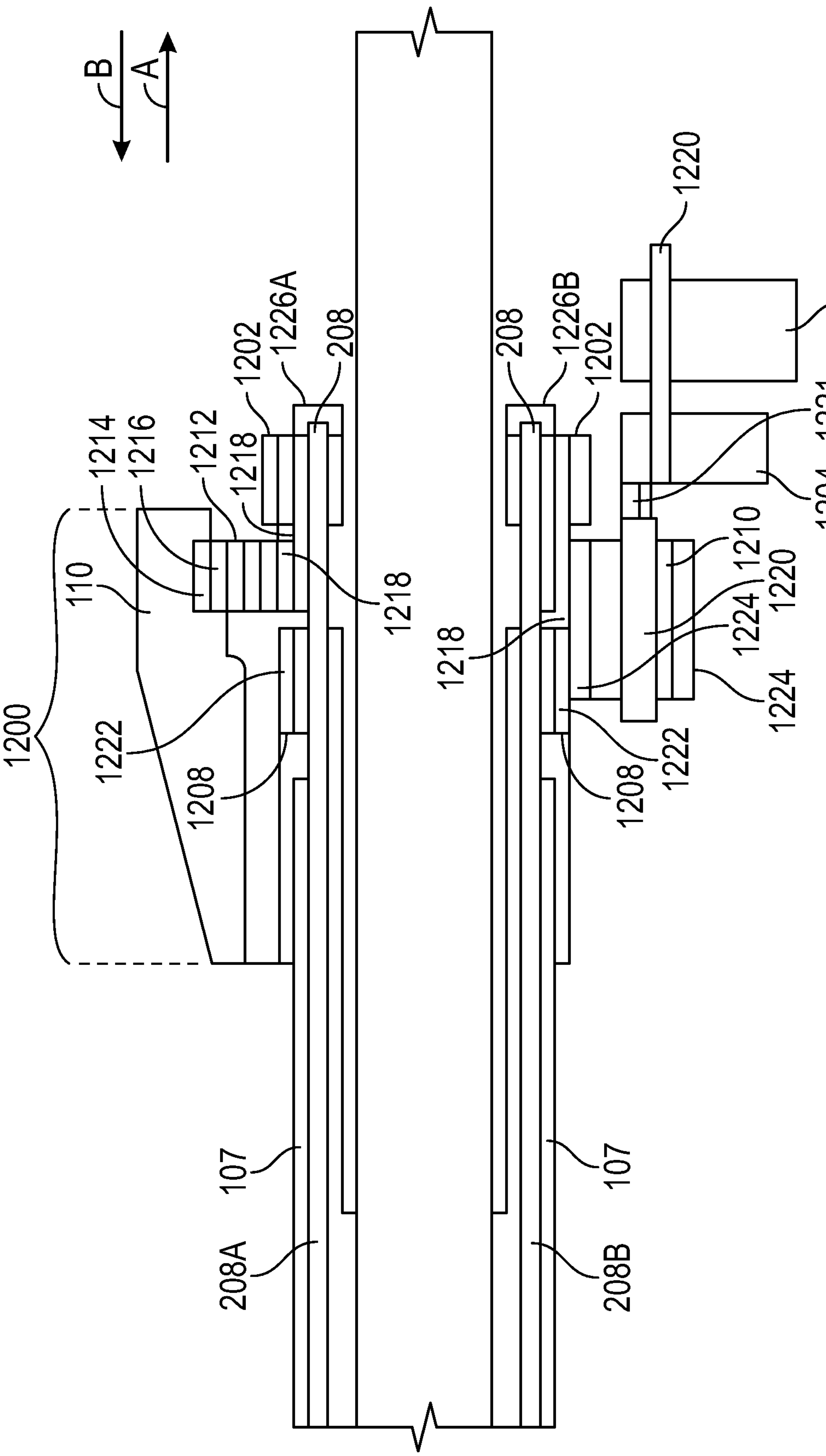
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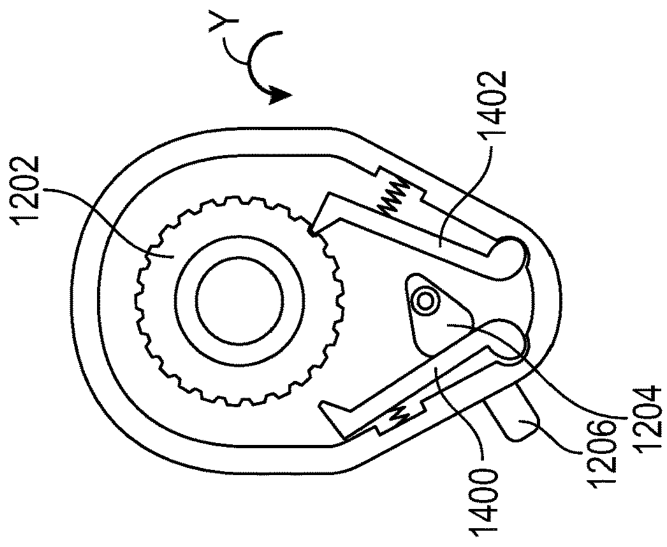
FIG. 12



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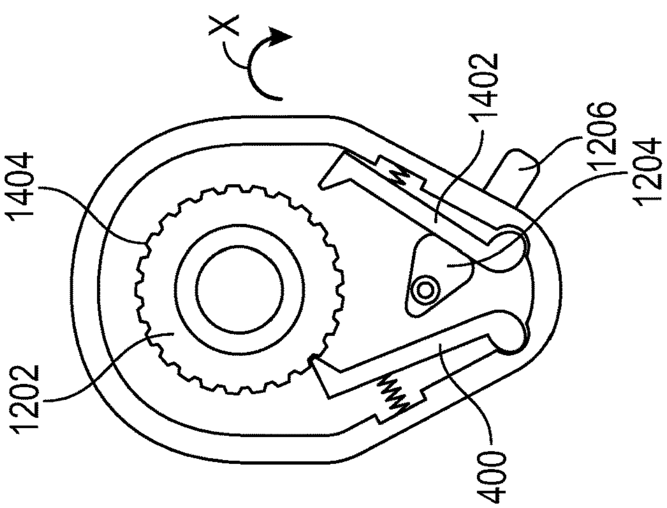
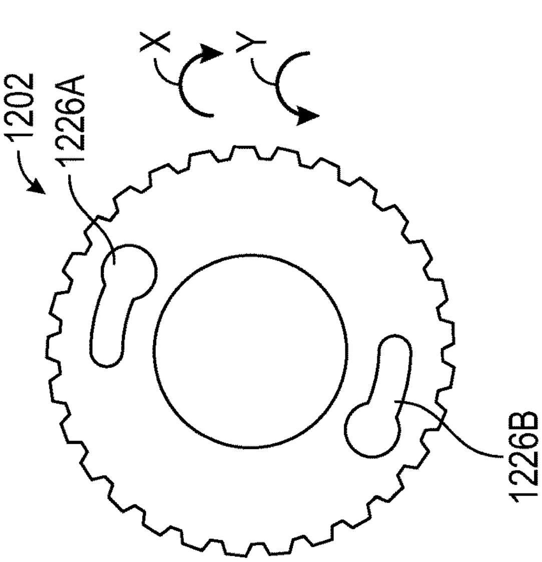


FIG. 14



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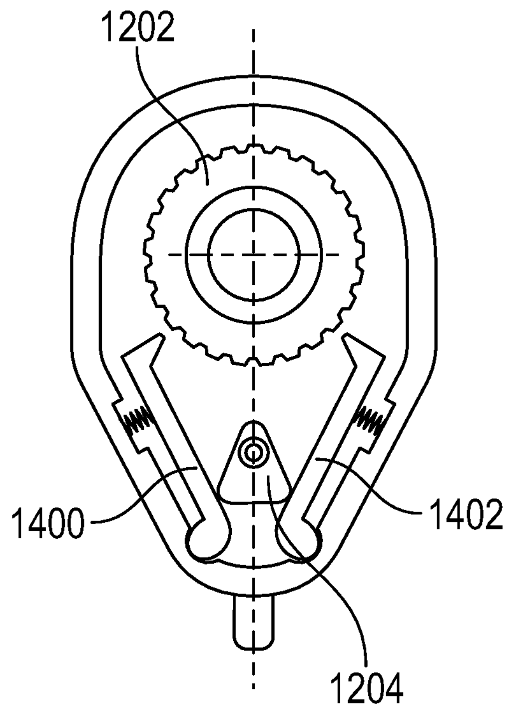

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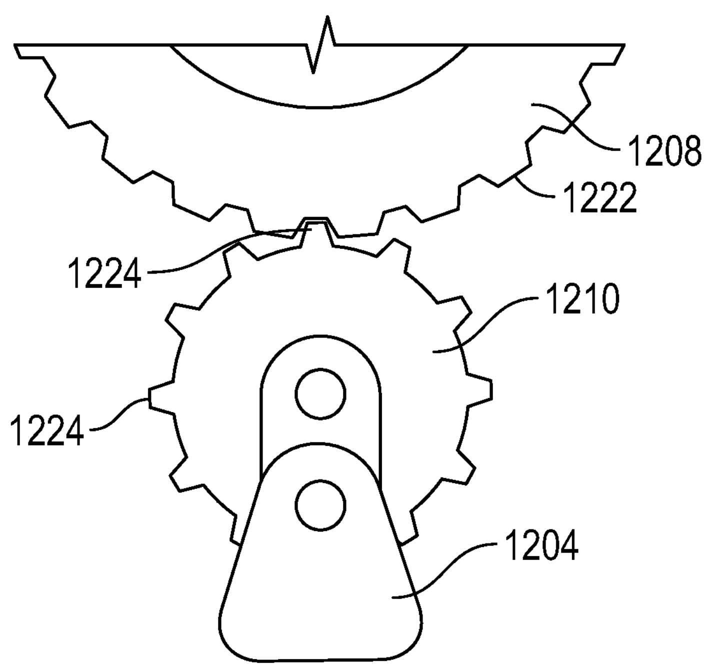

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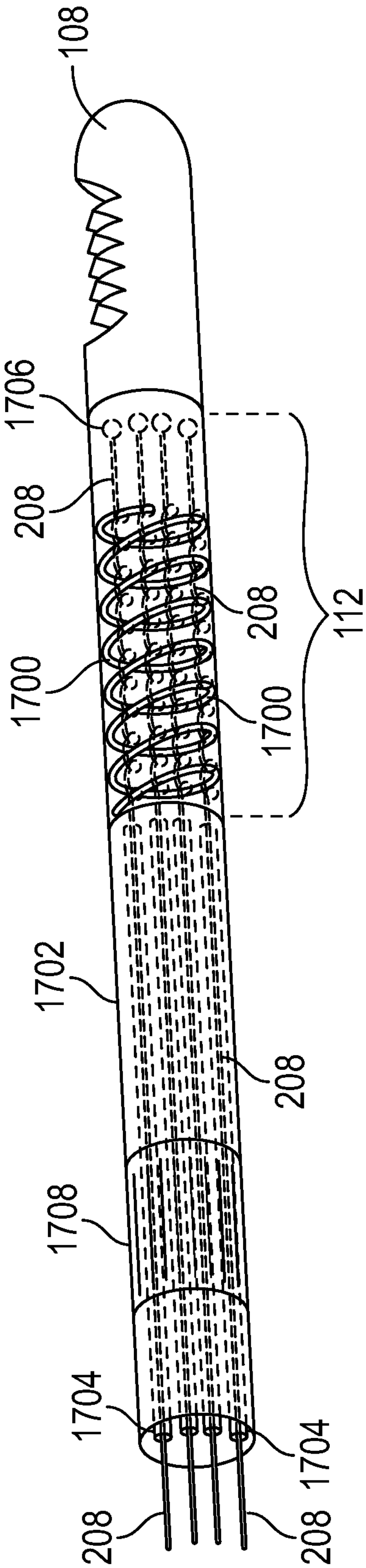

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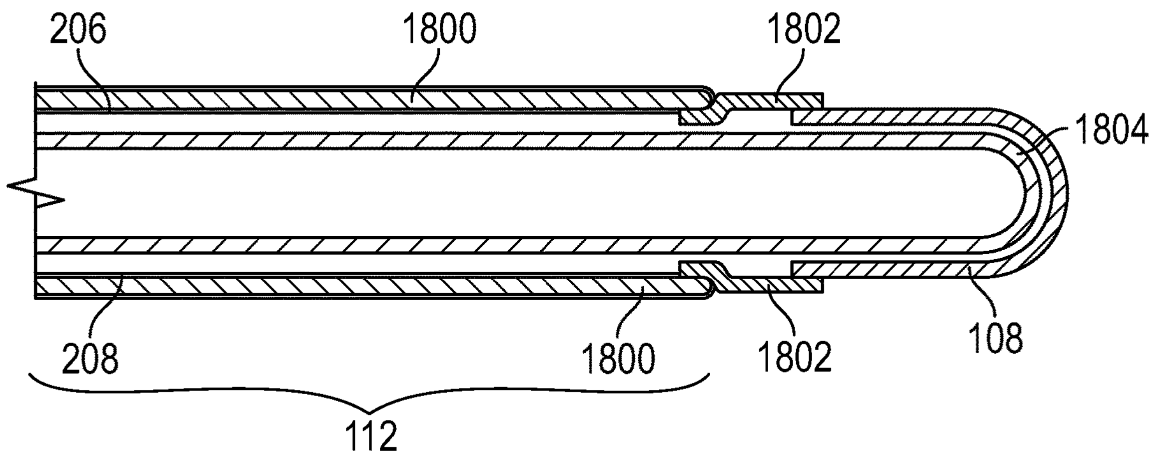

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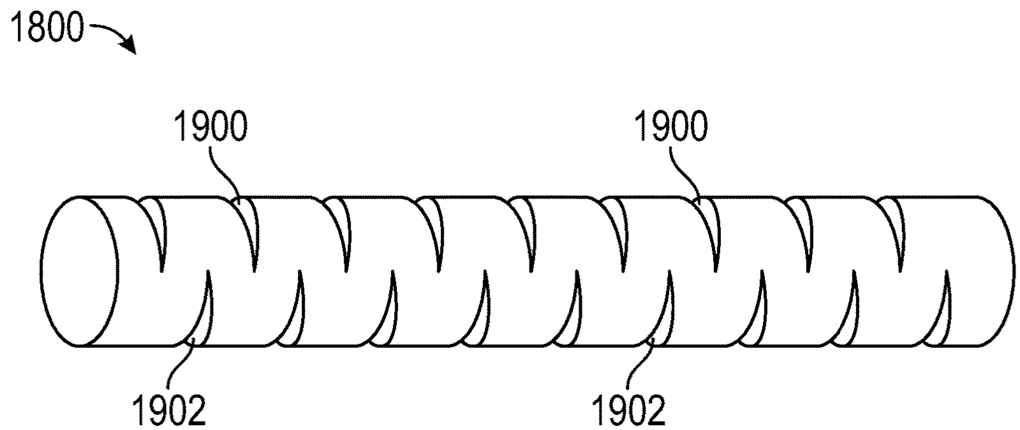

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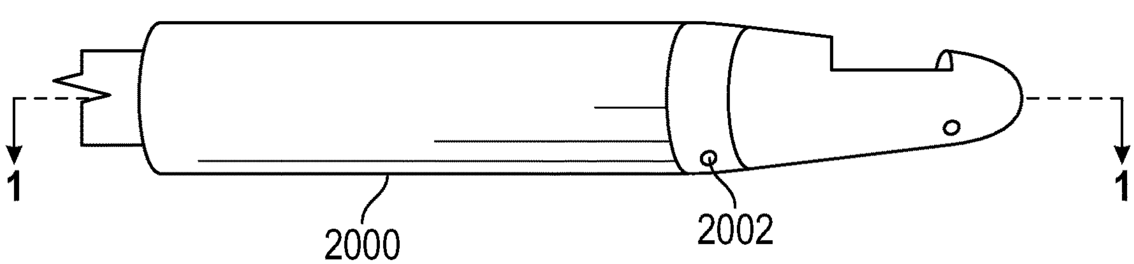

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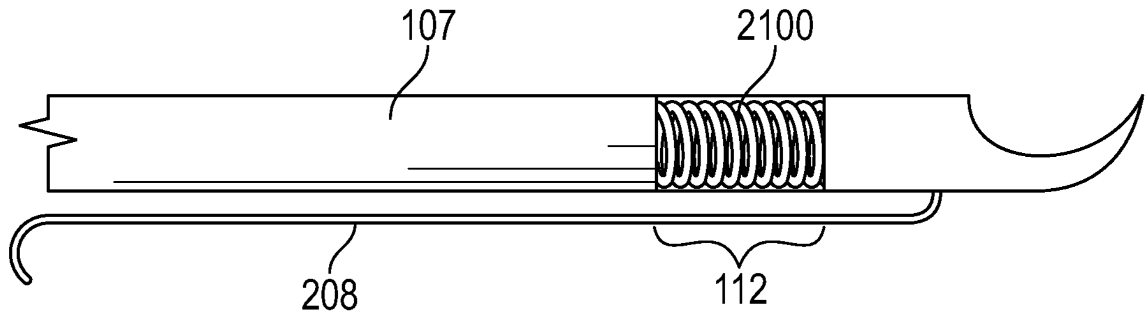

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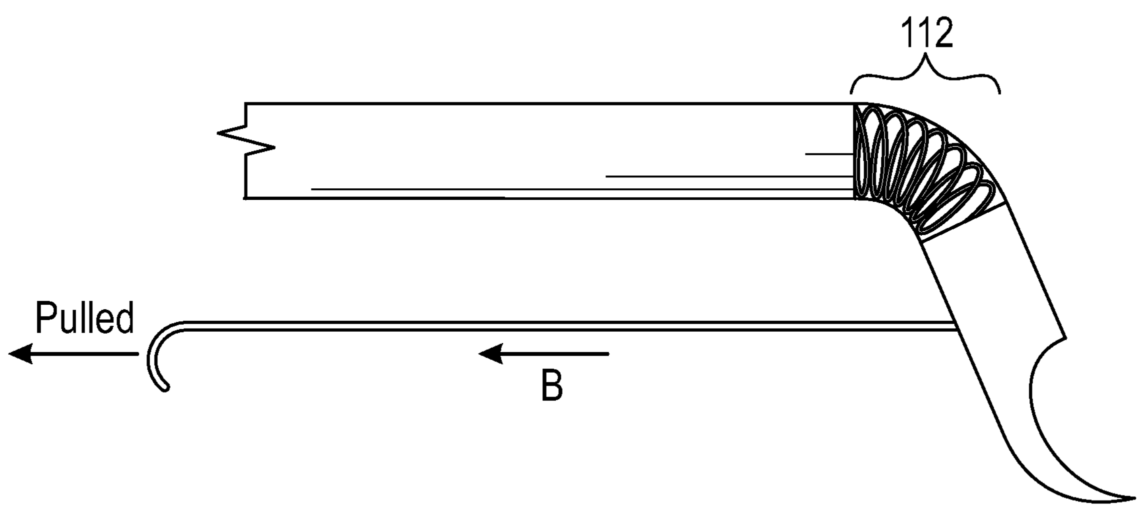

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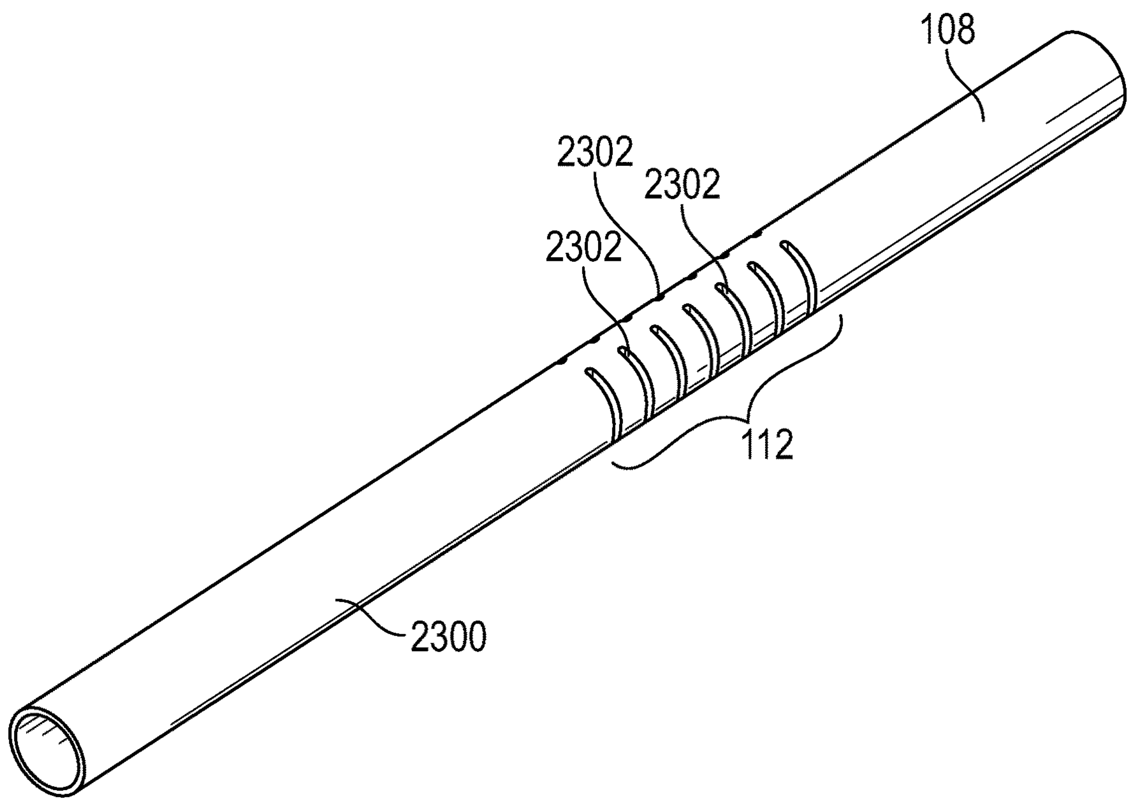

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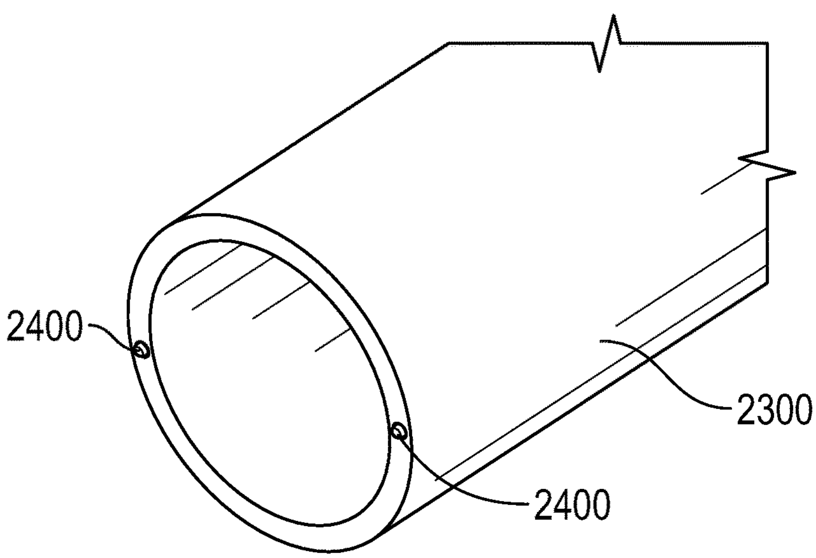

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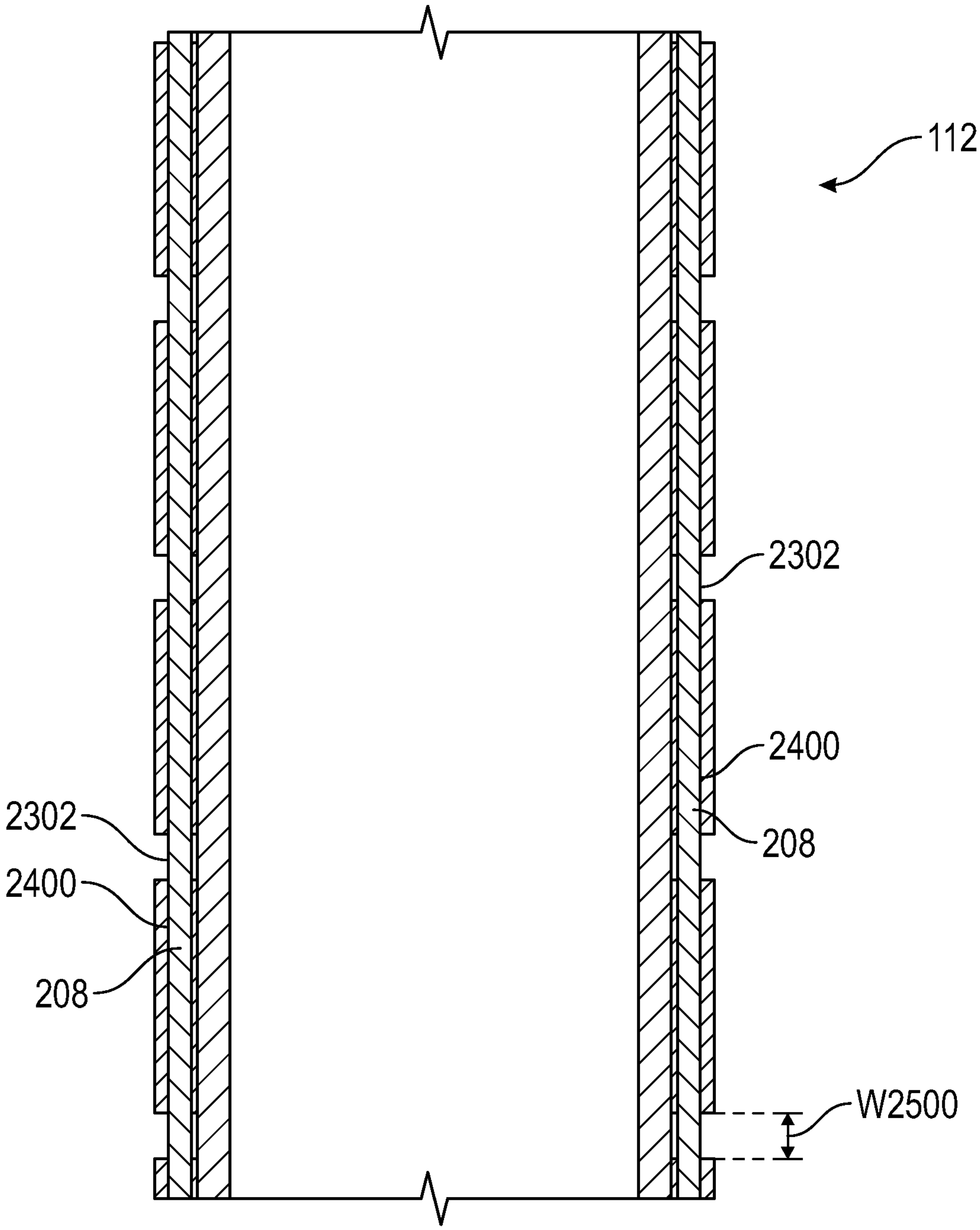

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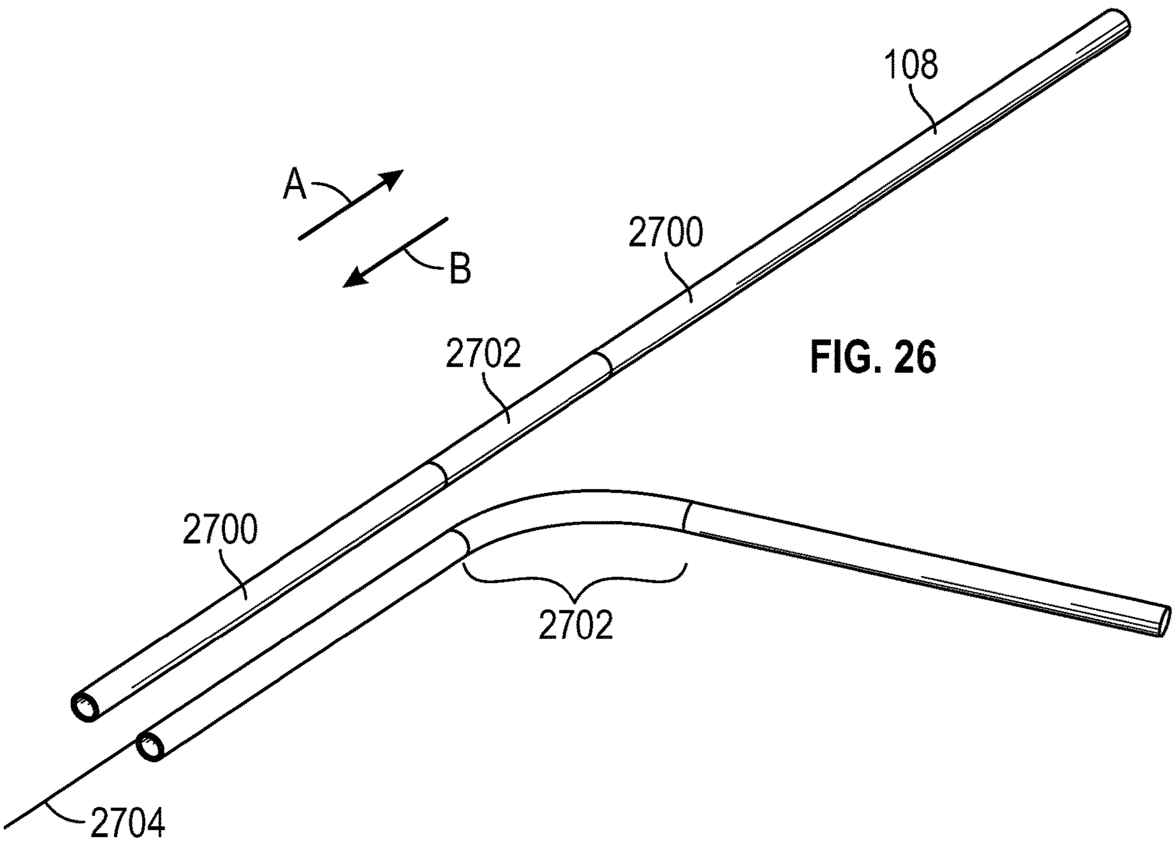
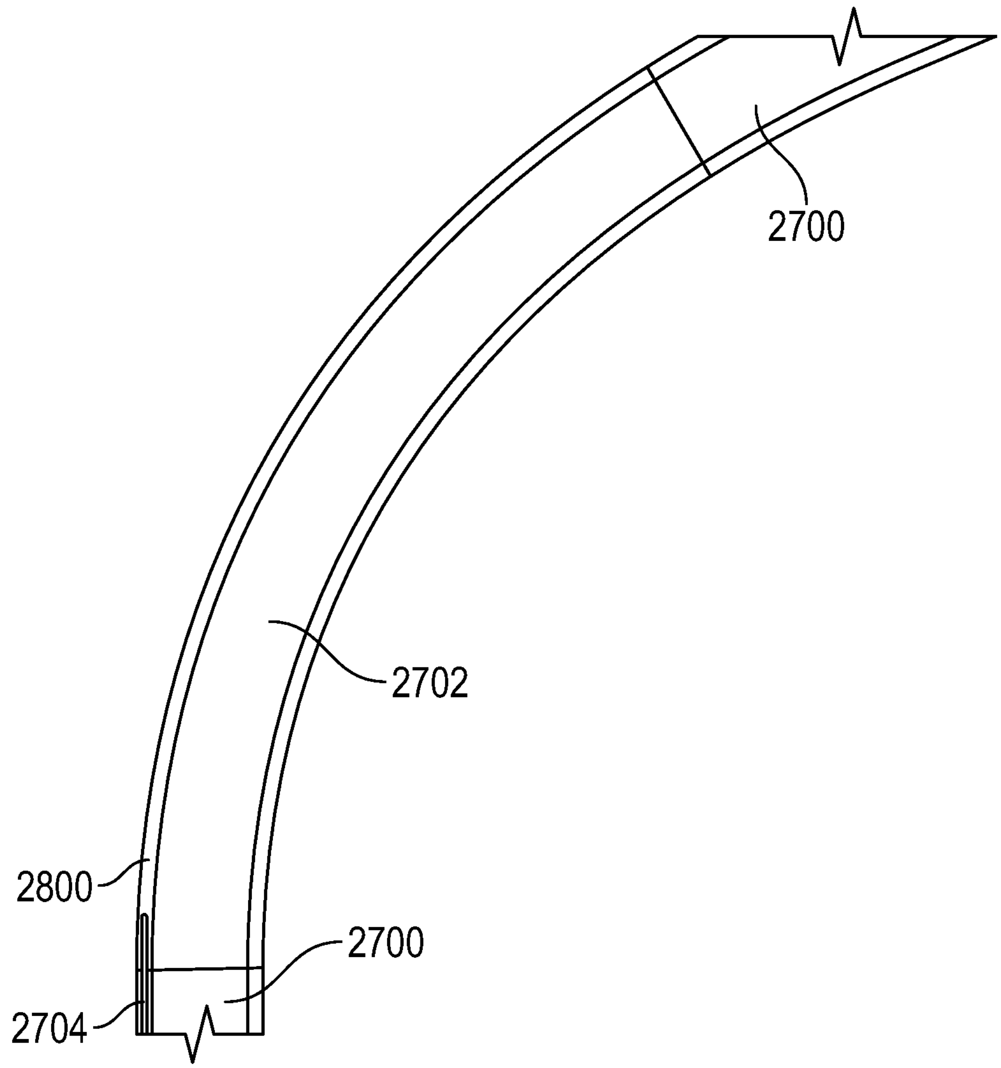


FIG. 26

FIG. 27



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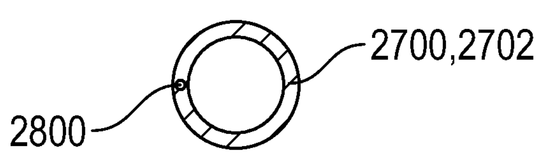

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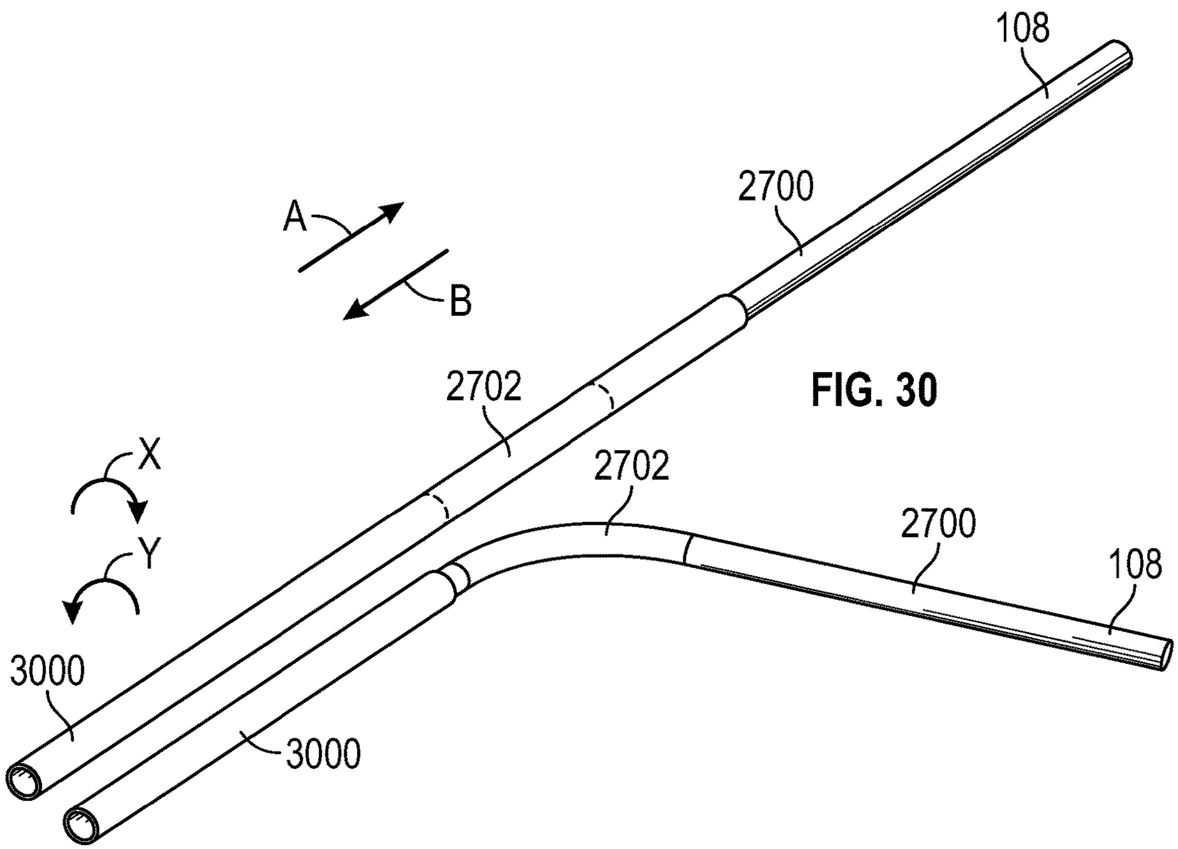


FIG. 30

FIG. 31

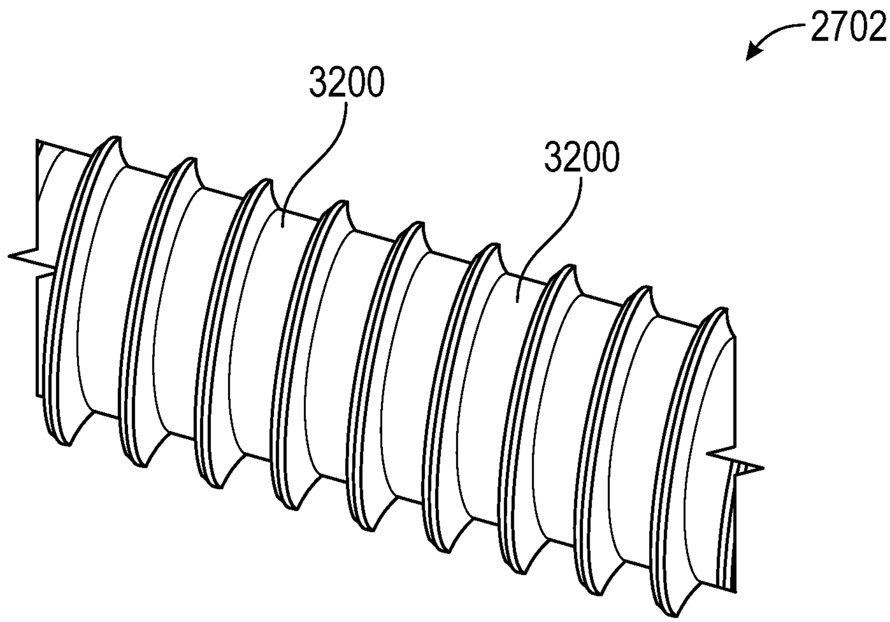
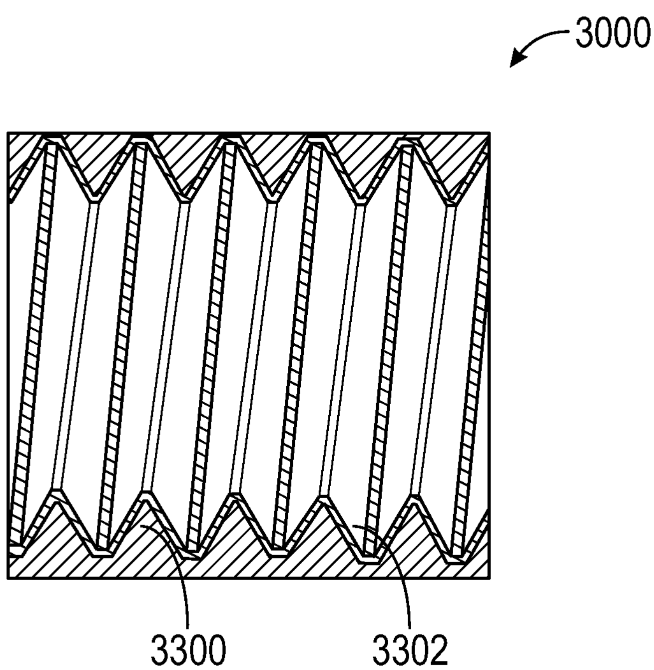


FIG. 32



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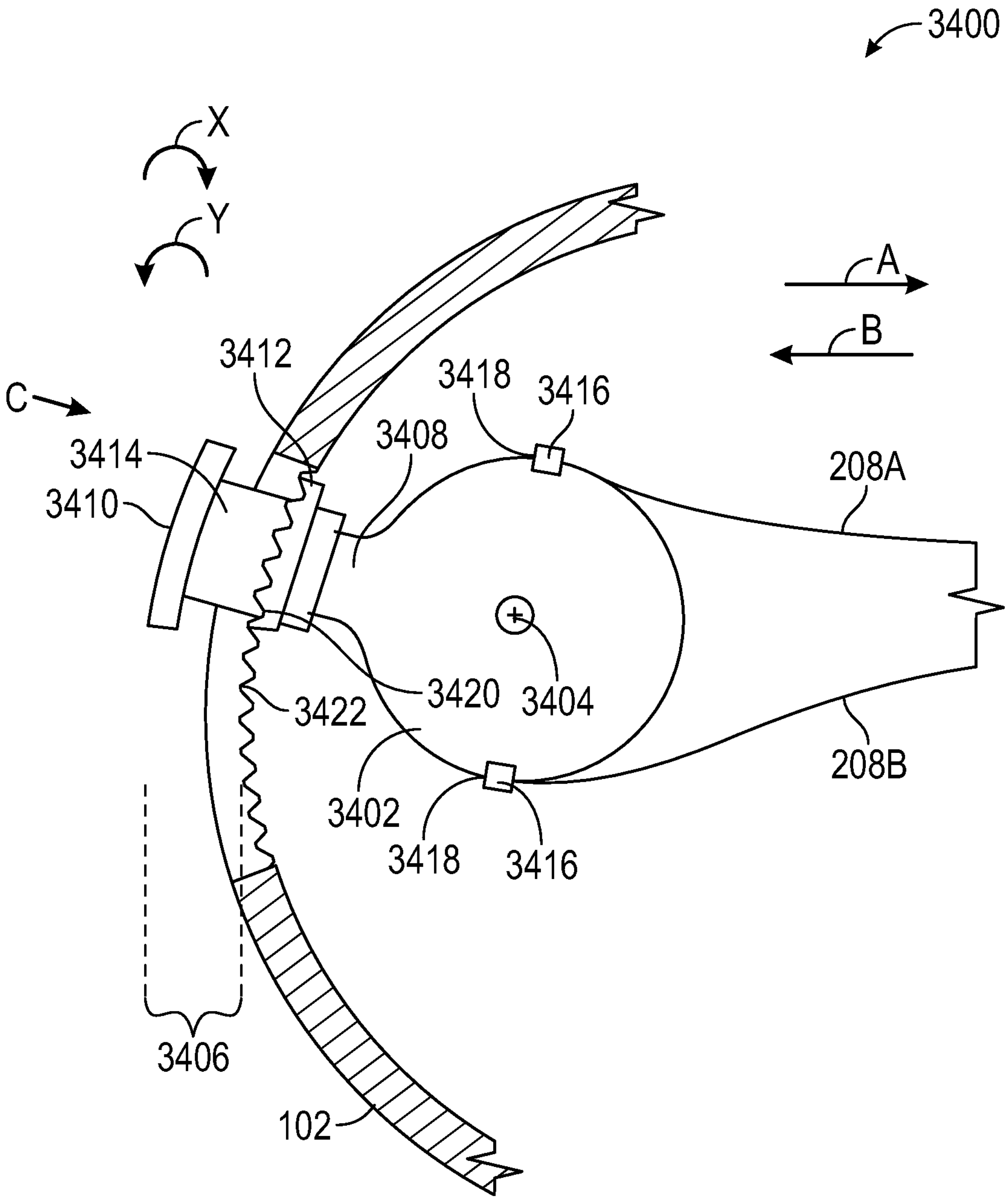
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FIG. 34

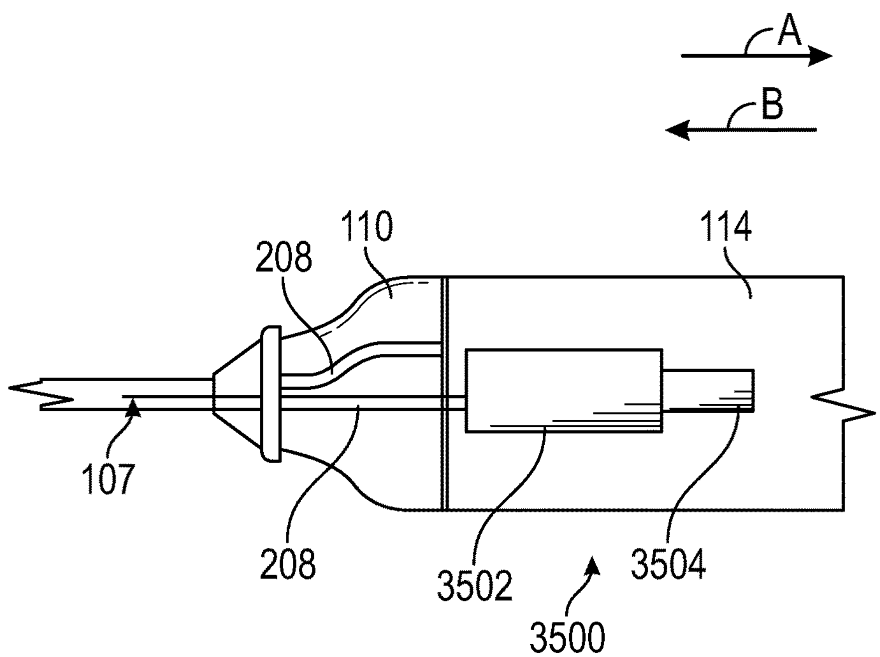


FIG. 35

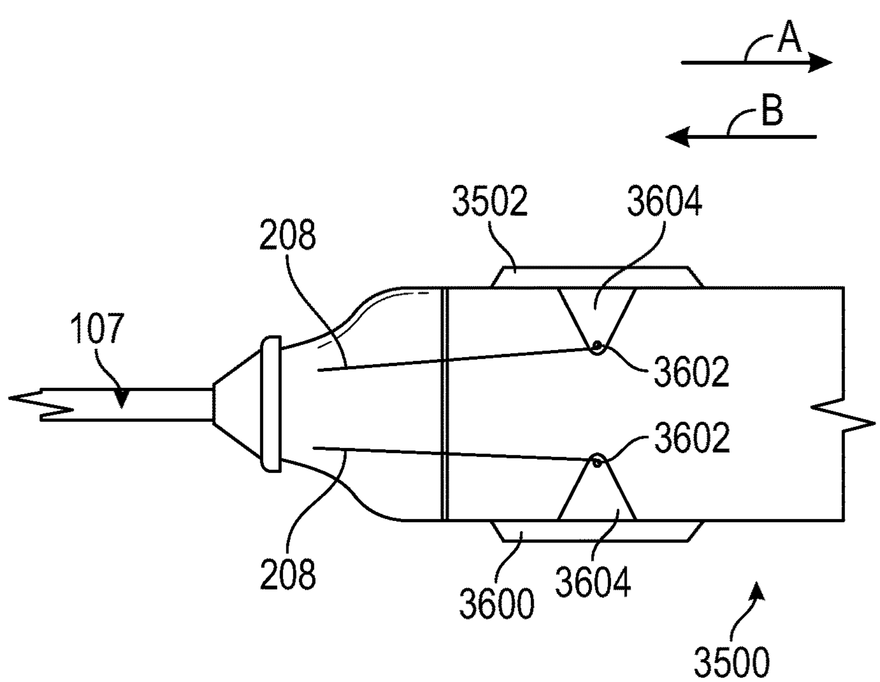
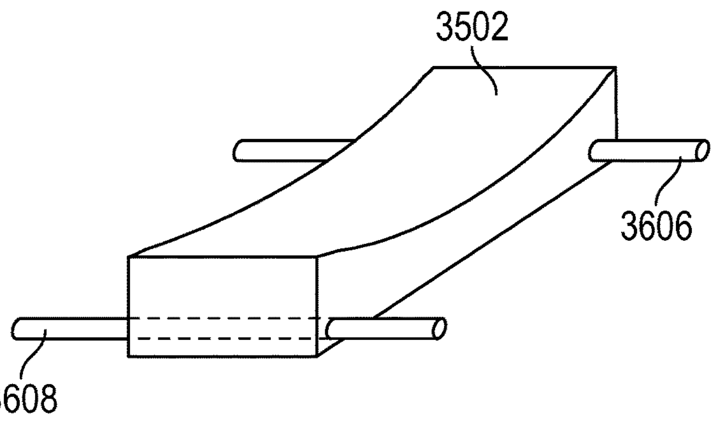


FIG. 36A



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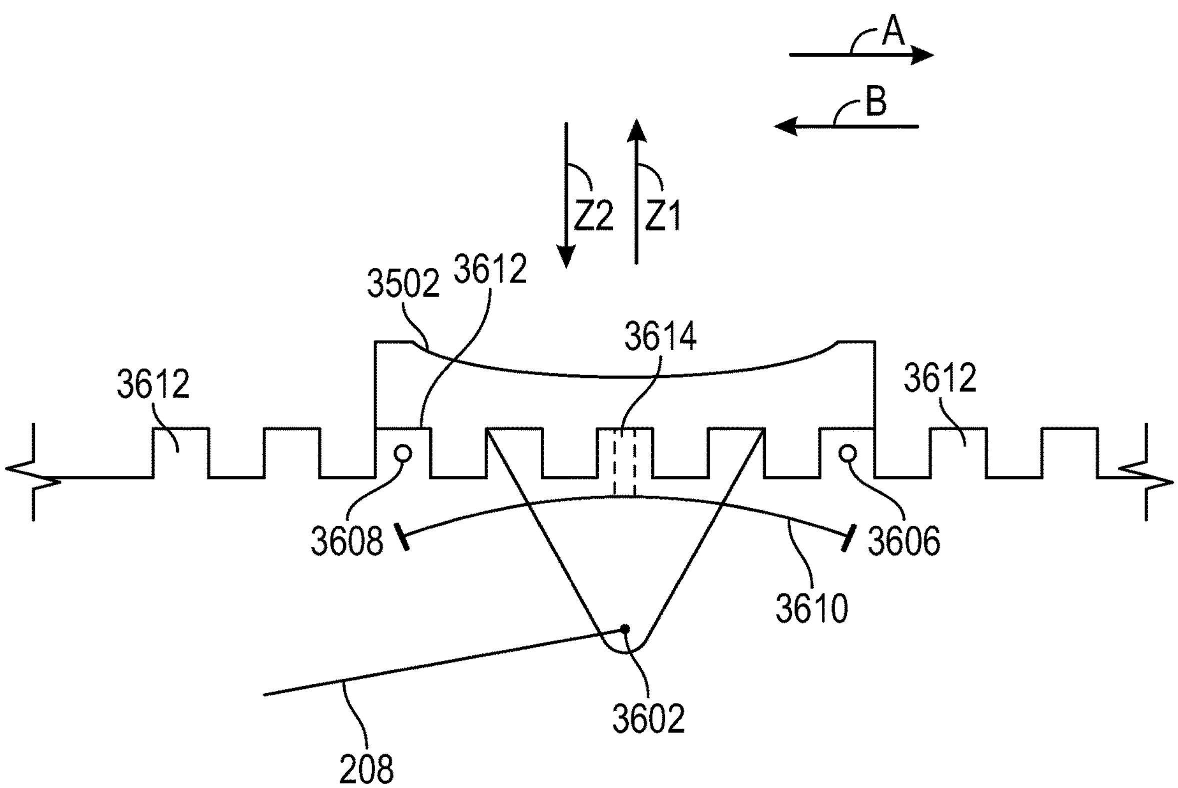

FIG. 36C

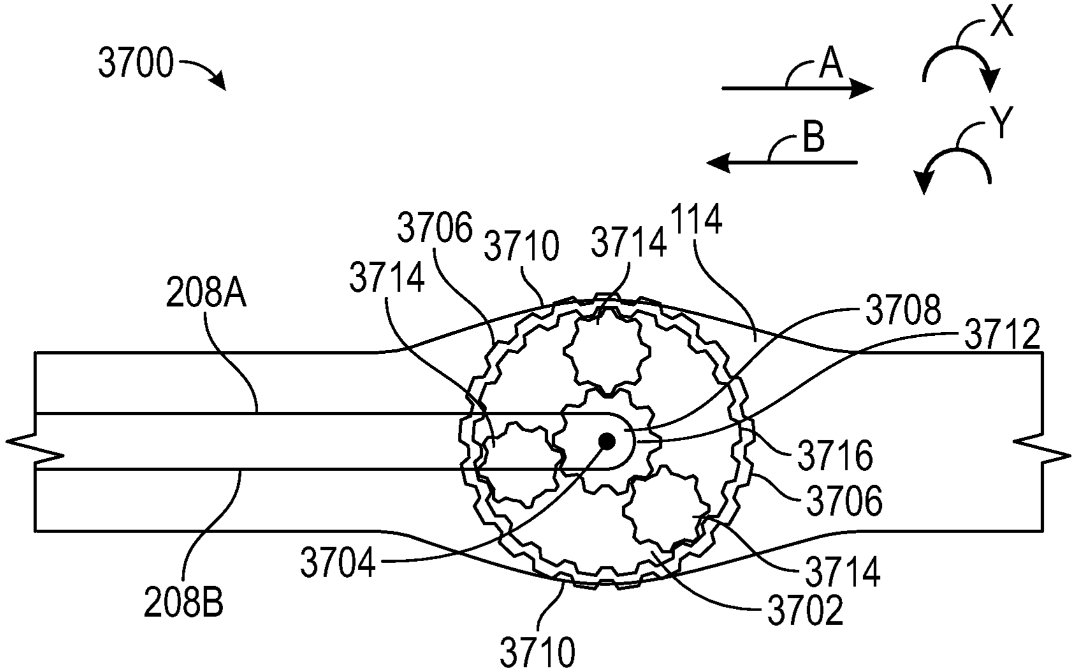

FIG. 37A

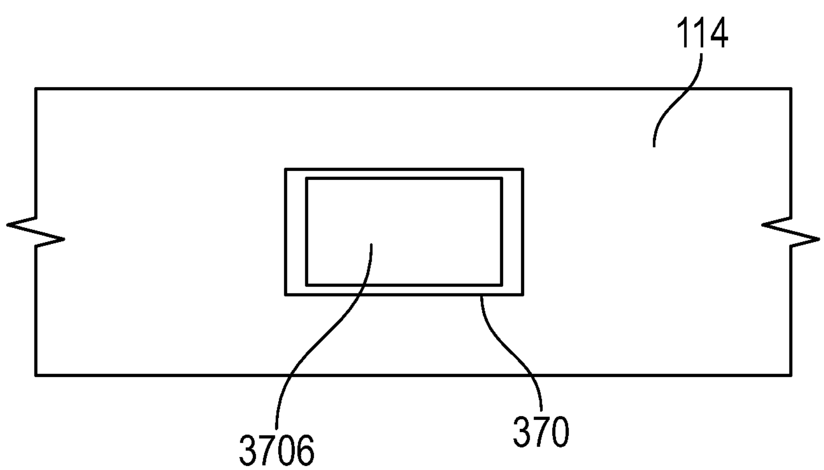

FIG. 37B

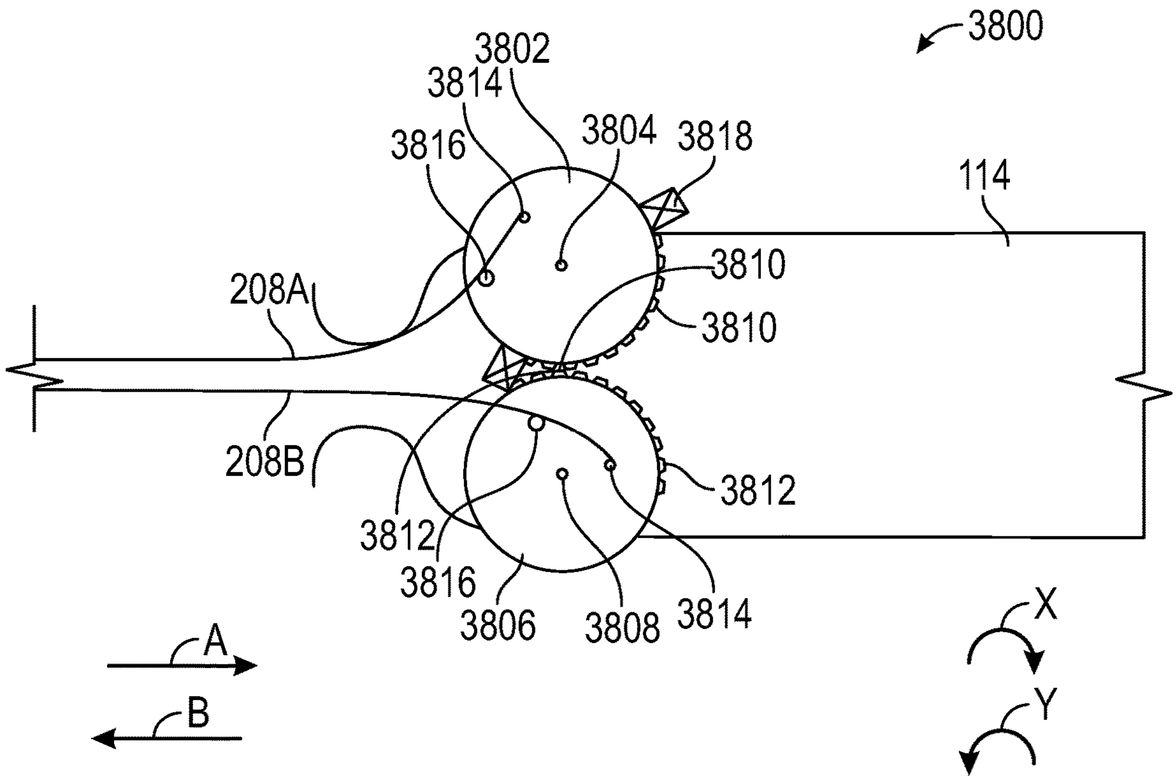

FIG. 38A

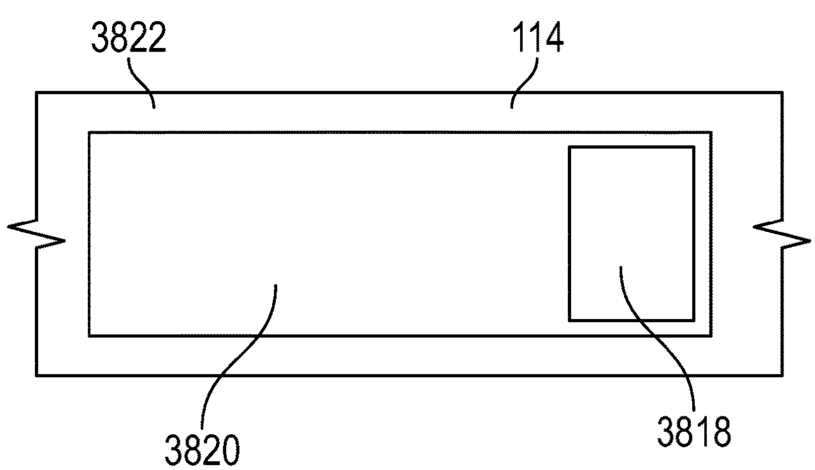

FIG. 38B

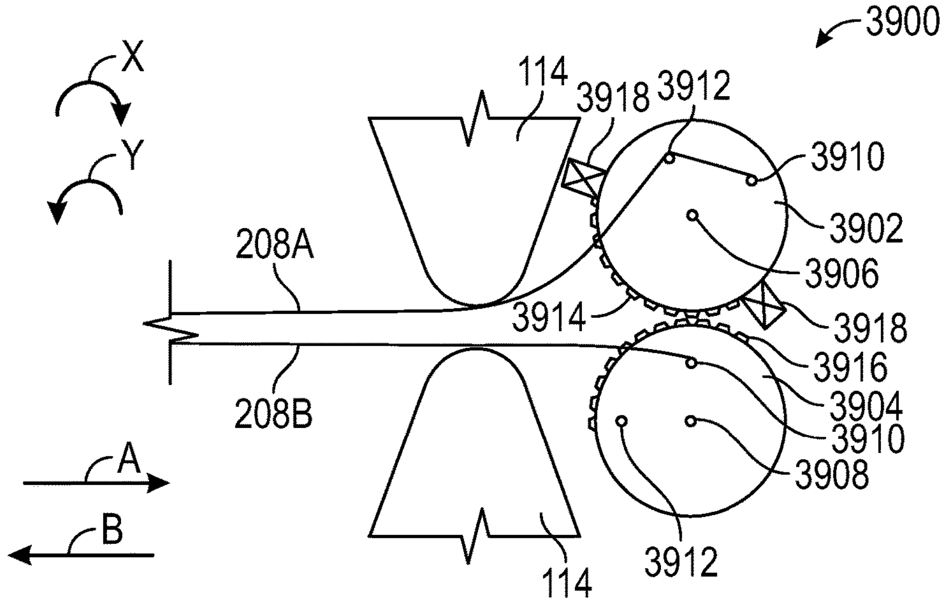
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FIG. 39

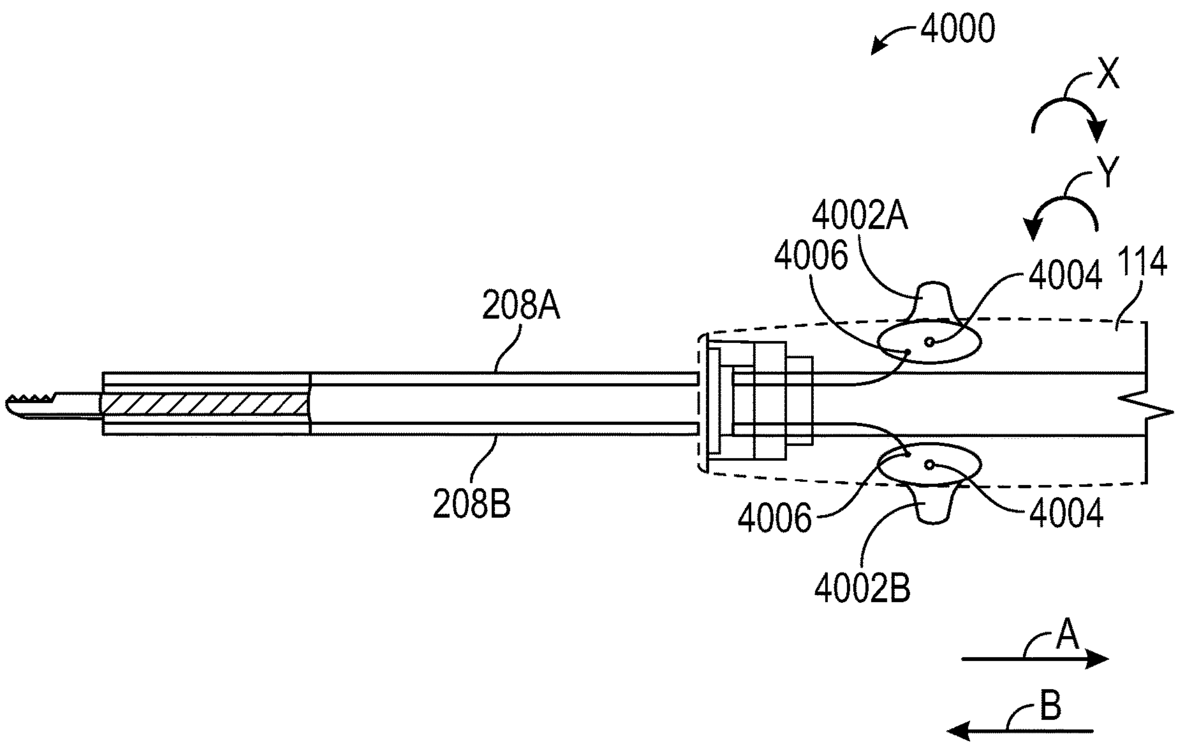
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FIG. 40

ARTICULATING DEBRIDER BLADE TIP AND HANDPIECE CONTROL

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of priority to U.S. Provisional Patent Application Ser. No. 63/203,733, filed Jul. 29, 2021 and U.S. Provisional Patent Application Ser. No. 63/203,735 filed Jul. 29, 2021, the contents of which are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

This document pertains generally, but not by way of limitation, to surgical devices that can be used for various surgical procedures. More specifically, but not by way of limitation, the present application relates to a handheld surgical instrument.

BACKGROUND

Occlusions within cavities of patients, such as sinus cavities, can cause a number of issues with a patient. These issues can include, for example, chronic rhinosinusitis, a deviated septum, nasal polyps, or the like. In order to remove these or other types of occlusions within cavities, a physician can use a handheld surgical instrument having a microdebrider or a drill. Microdebriders can be used with a variety of implements depending on the procedure being performed. A microdebrider can include a cutting implement that can oscillate, i.e., moves in a back and forth rotational motion or moves in a back and forth linear motion, and that can be used in Rhinologic procedures to remove softer tissues of the sinuses. For example, a pre-bent implement having cutting implements can be used when surgery is being performed at difficult to reach surgical sites, such as the aforementioned sinus cavities. In addition, straight implements can also be used to perform surgical procedures within sinus cavities. Moreover, cutting implements that can facilitate 360-degree rotation can also be used with a single microdebrider. Thus, a single microdebrider can be used for a variety of procedures with a variety of implements. As noted above, a drill can also be used. For example, a drill can be used in Otologic procedures to remove bone in, and around, the ear.

However, in situations where an implement is pre-bent, an angle at which the implement is pre-bent can limit how a microdebrider having the pre-bent implement can be used. For example, the implement can be pre-bent at an angle that does not allow the usage of the microdebrider within certain areas of a sinus cavity. Similarly, in situations where straight implements are used, the same problems may arise. In particular, the straight implement may not allow the usage of the microdebrider within certain areas of a sinus cavity.

Accordingly, what is needed is a handheld surgical device, such as a microdebrider or a drill, having an implement that can be bent by a user over a range of angles depending on the access requirements of a surgical site.

SUMMARY

Examples of the present disclosure relate to a handheld surgical instrument that facilitates articulation of a distal portion of the handheld surgical instrument while the distal portion is positioned at a surgical site. Articulation of the distal portion can be controlled at a handpiece of the handheld surgical instrument. The handheld surgical instrument can include a pull wire, such as a flat pull wire, that is anchored at the distal portion of the handheld surgical instrument and can extend towards the handpiece. The handpiece can include an assembly that can push and/or pull the flat pull wire. In an embodiment, the assembly can be controlled to push and/or pull the flat wire such that articulation of the distal portion can be controlled.

BRIEF DESCRIPTION OF FIGURES

FIG. **1** illustrates a handheld surgical instrument.

FIG. **2** illustrates an articulation portion of the handheld surgical instrument shown with reference to FIG. **1**.

FIGS. **3**A-**3**C show an articulation of a slit of the articulation portion shown with reference to FIG. **2**.

FIGS. **4**A and **4**B illustrate articulation of the articulation portion of FIG. **2**.

FIGS. **5**A-**5**C illustrate the functioning of slits of the articulation portion of FIG. **2**.

FIG. **6** illustrates a sectional view of a cutting portion of the handheld surgical instrument of FIG. **1**.

FIG. **7** is a top view of a cutting portion of the handheld surgical instrument of FIG. **1**.

FIGS. **8**A and **8**B show an articulation control assembly of the handheld surgical instrument of FIG. **1** in accordance with at least one example of the present disclosure.

FIG. **9** illustrates the coupling of an articulator of the articulation control assembly of FIGS. **8**A and **8**B with a spline in accordance with at least one example of the present disclosure.

FIG. **10** is a top view of the articulation control assembly of FIGS. **8**A and **8**B in accordance with at least one example of the present disclosure.

FIGS. **11**A and **11**B illustrate the use of a handle gear in conjunction with the articulation control assembly of FIGS. **8**A and **8**B to control articulation of a distal end of the handheld surgical instrument of FIG. **1** in accordance with at least one example of the present disclosure.

FIG. **12** shows an articulation control assembly of the handheld surgical instrument of FIG. **1** in accordance with at least one example of the present disclosure.

FIGS. **13**-**16**B illustrate a functioning of a ratchet assembly of the articulation control assembly of FIG. **12** in accordance with at least one example of the present disclosure.

FIGS. **17**-**25** illustrate alternative embodiments of the articulation portion of FIG. **2** in accordance with at least one example of the present disclosure.

FIGS. **26**-**33** illustrate alternative embodiments of the articulation portion of FIG. **2** in accordance with at least one example of the present disclosure.

FIGS. **34**-**40** show alternative embodiments of articulation control assemblies that may be used with the handheld surgical instrument of FIG. **1** in accordance with at least one example of the present disclosure.

DETAILED DESCRIPTION

Examples of the present disclosure relate to a handheld surgical instrument that facilitates articulation of a distal portion of the handheld surgical instrument while the distal portion is positioned at a surgical site. Articulation of the distal portion can be controlled at a handpiece of the handheld surgical instrument. In an embodiment, the handheld surgical instrument can include an outer blade which can remain stationary during a surgical procedure and an inner blade disposed within the outer blade that can rotate within the outer blade during operation of the handheld surgical instrument. A distal portion of the outer blade can include opposing slits, which facilitates articulation in a first direction such that a tip of the handheld surgical instrument can have a bend radius up to two inches while articulating up to ninety degrees. The slits can be configured such that the distal portion only articulates in the first direction.

In an embodiment, articulation of the distal portion can be controlled at the handpiece of the handheld surgical instrument. The handheld surgical instrument can include a pull wire, such as a flat pull wire, that is anchored at the distal portion of the handheld surgical instrument and can extend towards the handpiece. The handpiece can include an assembly that can push and/or pull the pull wire. In an embodiment, when the assembly is actuated to pull the pull wire, the distal portion can articulate in order to bend the distal portion of the handheld surgical instrument. Moreover, in an embodiment, the assembly can be actuated to push the pull wire, thereby straightening the distal portion of the handheld surgical instrument.

Now making reference to the Figures, and more specifically FIG. **1**, a handheld surgical instrument **100** is shown. The handheld surgical instrument **100** can include a handpiece **102**, a handle gear **104**, and an actuating mechanism **106**. The handheld surgical instrument **100** can also include a shaft **107** that extends from a proximal end of the handheld surgical instrument **100** to a distal end of the handheld surgical instrument **100** where a cutting implement **108** is disposed at a distal end of the shaft **107**. In an example, the handheld surgical instrument **100** can be a microdebrider that can be used to treat various rhinological conditions, such as chronic rhinosinusitis, a deviated septum, nasal polyps, or the like. While not shown, the handheld surgical instrument **100** can include a rotating blade disposed within the cutting implement **108** that extends from the cutting implement **108** to the proximal end of the handheld surgical instrument. the rotating blade can work in conjunction with the cutting implement **108** to resect tissue during use of the handheld surgical instrument **100**. Therefore, the cutting implement **108** can function as an outer blade and the rotating blade can function as an inner blade that rotates within the cutting implement **108**.

The cutting implement **108** can be any type of blade that can be used for various procedures, such as functional endoscopic sinus surgery (FESS) procedures. The type of blades that can be used for the cutting implement **108** can include a standard blade with or without suction capability, bipolar and/or monopolar energy blades that can be used for hemostasis during a surgical procedure, or a rotatable blade. Moreover, the cutting implement **108** can be a straight blade, an angled blade, or a turbinate blade. In addition, while the cutting implement **108** is shown, the handheld surgical instrument **100** can also include a burr in place of the cutting implement **108** for FESS procedures, such as a diamond bullet burr, a diamond ball burr, a diamond tapered burr, a fluted barrel burr, or an angled burr. In addition, the handheld surgical instrument **100** can be used for procedures other than FESS procedures. Moreover, in further examples, the handheld surgical instrument **100** can be used as a dental drill, as an ontology drill for ENT applications, or with a Dremel™. The cutting implement **108** can be rotated with a nose cone **110** and can include an articulation portion **112** along with a housing **114** adjacent the nose cone **110**. It should be noted that in some embodiments, articulation may only occur in one plane.

The articulation portion **112** can be configured to allow articulation of the cutting implement. The articulation portion can be configured to allow a user to articulate the distal end of the handheld surgical instrument **100**. For example, the user can be a surgeon and when the surgeon maneuvers the cutting implement **108** and the distal end of the handheld surgical instrument **100** to a surgical site, the surgeon can articulate the distal end of the handheld surgical instrument **100**. By articulation of the distal end, the cutting implement **108** can be properly positioned at the surgical site and the surgeon can resect tissue.

In order to allow articulation of the cutting implement **108**, the articulation portion **112** can be formed of any material that allows for bending of the articulation portion **112**. Examples of materials that can be used can include a soft metal, polyurethane, a plastic, or the like. Regardless of the material used to form the articulation portion **112**, in an example, as shown with reference to FIG. **2**, the articulation portion **112** can include slits **200** and **202** that can be disposed opposite to each other at the articulation portion **112**. Moreover, as may be seen with regards to FIG. **2**, the slits **200** can be offset from the slits **202**. The slits **200** and **202** can be configured to allow articulation of the articulation portion **112** in a direction X and a direction Y. In particular, the slit **200** can include a flex region **204** which can allow for flexing of the slit **200** during articulation of the articulation portion **112**. More specifically, the flex region **204** can allow side walls of the slit **200** to rotate about the directions X and Y, thereby allowing articulation of the slit **200** and the articulation portion **112**.

To further illustrate, making reference to FIGS. **3A**-**3C**, which show the articulation of the slit **200**, the slit **200** can include side walls **300** and **302** which can be separated by a distance W_{304} from each other. During articulation of the articulation portion **112**, the side wall **300** can move along the direction X and the side wall **302** can move along the direction Y such that the side walls **300** and **302** can move closer to each other and can have a distance W_{306} therebetween, as shown with reference to FIG. **3**B. Here, the articulation portion **112** can be articulated such that the cutting implement **108** can have the configuration shown with respect to FIG. **4**A.

Additionally, should more articulation be required, the side wall **300** can move further along the direction X and the side wall **302** can move further along the direction Y such that the side wall **300** moves closer to the side wall **302** and can be separated by a distance W_{308}, as shown with reference to FIG. **3**C. The articulation portion **112** can be articulated such that the cutting implement **108** can have the configuration shown with respect to FIG. **4**B. As discussed herein, articulation of the cutting implement **108** from a substantially flat position to the position shown with reference to FIG. **4**A can be described as positive articulation. Moreover, articulation of the cutting implement **108** from the position shown with reference to FIG. **4**A to the position shown with reference to FIG. **4**B can be also be described as positive articulation. Articulation of the cutting implement **108** from the position shown with reference to FIG. **4**B to the position shown with reference to FIG. **4**A can be described as negative articulation. In addition, articulation of the cutting implement **108** from the position shown with reference to either FIG. **4**B or FIG. **4**A to a substantially flat position can be described as negative articulation. As may be seen with reference to FIG. **4**B, the articulation portion **112** can be configured such that the distal end of the cutting implement **108** can have a bend radius B_R that is up to two inches. The bend radius B_R can be in a range of about 0.75 to about 1.25 inches.

Returning attention to FIG. **2**, the slit **202** can include a flex region **206** which can allow for flexing of the slit **202** during articulation of the articulation portion **112**. More specifically, the flex region **206** can rotate about the directions X and Y, thereby allowing articulation of the slit **200** and the articulation portion **112**. The flex region **206** can allow side walls of the slit **202** to rotate about the directions X and Y, thereby allowing articulation of the slit **202** and the articulation portion **112**.

To further illustrate, making references to FIGS. **5**A-**5**C, the slit **202** can include side walls **500** and **502** which can be separated a distance W_{504} from each other. During articulation of the articulation portion **112**, the side wall **500** can move along the direction X that the side walls **500** and **502** can move further apart and can have a distance W_{506} therebetween, as shown with reference to FIG. **5**B. The articulation portion **112** can be articulated such that the cutting implement **108** can have the configuration shown with respect to FIG. **4**A.

Additionally, should more articulation, such as additional positive articulation, be required, the side wall **500** can move further along the direction X such that the side wall **500** moves further away from the side wall **502** and can be separated by a distance W_{508}, as shown with reference to FIG. **5**C. The articulation portion **112** can be articulated such that the cutting implement **108** can have the configuration shown with respect to FIG. **4**B. It should be noted that as the side wall **500** moves as discussed with reference to FIGS. **5**A-**5**C, the side wall **500** can simultaneously move with the side walls **300** and **302**, as discussed above with regards to FIGS. **3**A-**3**C.

Returning reference to FIG. **2**, in order to control the articulation of the articulation portion **112**, the handheld surgical instrument **100** can include flat pull wires **208** that can be disposed opposite each other on the cutting implement **108**. The flat pull wires **208** can be formed of any material that allows for articulation of the flat pull wires **208**. Examples of materials that can be used can include a soft metal, polyurethane, a plastic, or the like. As shown with reference to FIG. **2**, the flat pull wires **208** can be disposed at the articulation portion **112** at the slits **200** and **212**. The cutting implement **108** can include passageways **210** through which the flat pull wire **208** can pass in the cutting implement **108**, as shown with reference to FIGS. **2** and **6**. Furthermore, the articulation portion **112** can include areas **212** within which the flat pull wires **208** are exposed. Articulation of the articulation portion **112** can occur at the cutting implement areas **212**.

The flat pull wires **208** can anchor with the cutting implement **108** at anchor points **700** as shown with reference to FIG. **7**. The flat pull wires **208** can anchor with the cutting implement **108** at the anchor points **700** at an anchoring area **702** with soldering, spot welding, threaded fasteners, rivets, or the like. Furthermore, the flat pull wires **208** can be anchored to the cutting implement **108** at the anchoring area with an adhesive, such as an epoxy, a glue, or the like. One of the flat pull wires **208** is anchored at the anchoring area **702** located adjacent to a cutting window of the cutting implement **108**. In other words, the cutting window and one of the flat pull wires **208** are disposed on same side of the shaft **107**. The flat pull wires **208** can have a width W_{704} that can be in a range of about 0.06 inches to about 0.18 inches. However, the width W_{704} can vary depending on a thickness of the flat pull wires **208** that can accommodate a given flexibility and tensile strength requirement. The cutting implement passageway can have a width W_{600} that can be in a range of about 0.08 inches to about 0.20 inches. In addition to having a flat configuration as shown with reference to FIGS. **2** and **7**, the flat pull wire **208** can have a circular configuration where a cross-section of the flat pull wire **208** is circular, an oval configuration where a cross-section of the flat pull wire **208** is in the shape of an oval, or any other type of configuration. Moreover, the cutting implement passageway **210** can have a configuration that complements the flat pull wire **208**. Thus, if the pull wire **208** has a circular or oval configuration, the cutting implement passageway **210** can have a circular or oval configuration. Moreover, the flat pull wires **208** can be formed of any semi-rigid material. Examples of materials that can be used for the flat pull wires **208** can include a bendable metal, such as aluminum, stainless steel, galvanized steel, plastics, or the like.

As noted above, a user, such as a surgeon, can articulate the distal end of the handheld surgical instrument **100** via the articulation portion **112** such that the cutting implement **108** can be properly positioned at a surgical site. The user can articulate the distal end of the handheld instrument **100** along with the cutting implement **108** at the handpiece **102**. The handpiece **102** can include an articulation control assembly **800** that can be used to articulate the distal portion of the handheld surgical instrument **100** and the cutting implement **108**, as shown with reference to FIG. **8**A. The articulation control assembly **800** can include the actuating mechanism **106** along with an actuating mechanism **802** in communication with the actuating mechanism **106** via a shaft **828**.

The articulation control assembly **800** can include a housing **804** that can include portions **804**A and **804**B. The actuating mechanism **106** can be coupled to a spur gear **806** of the articulation control assembly **800** where the spur gear **806** can be disposed within the articulation control assembly housing portion **804**A. The actuating mechanism **802** can be coupled to a spur gear **808** of the articulation control assembly **800**. In an embodiment, the spur gear **808** can be disposed within the articulation control assembly housing portion **804**B.

In an embodiment, the handle gear **104** can include a spline **810**, which can operatively couple with the spur gear **806**. The handle gear **104** can also include a spline **812**, which can operatively couple with the spur gear **808**. Moreover, the articulation control assembly **800** can include a one way bearing **814** disposed within a recess **816** of the spur gear **806** along with another one way bearing **818** disposed within a recess **820** of the spur gear **808**.

In an embodiment, the articulation control assembly **800** can include splines **822** and **824**, which, in an embodiment, can be configured to operatively couple with an articulator **826**. In an embodiment, the articulator **826** can be a gear and can include gear teeth **900** that are complimentary to gear teeth **902** of the spline **822**, a shown with reference to FIG. **9**. In addition, the articulator **826** can also include recesses **827** within which an end **1000** (FIG. **10**) of the flat pull wire **208** can be disposed. In an embodiment, the pull wire end **1000** anchors to the pull wire gear recess **827**. In an embodiment, the pull wire end **1000** can be anchored within the pull wire gear recess **827** with soldering, spot welding, threaded fasteners, rivets, or the like. Furthermore, the pull wire end **1000** can be anchored within the pull wire gear recess **827** with an adhesive, such as an epoxy, a glue, or the like.

Returning attention to FIG. **8**A, the articulation control assembly **800** can include the shaft **828** having ends **828**A and **828**B. The actuating mechanism **106** and the spur gear **806** can be disposed at the shaft end **828**A while the actuating mechanism **802** and the spur gear **808** can be disposed at the shaft end **828**B. When a user engages the actuating mechanism **106** such that the actuating mechanism moves along a direction A, the actuating mechanism **802** along with the spur gears **806** and **808** can move along the direction A. In addition, when a user engages the actuating mechanism **106** such that the actuating mechanism moves along the direction A, the splines **822** and **824** can also move along the direction A. In an embodiment, the spur gears **806** and **808** along with the splines **822** and **824** can continue to move along the direction A until an end **830** of the spur gear **808** nears a surface **832** of the articulation control assembly housing portion **804**B. As may be seen with reference to FIG. **8**A, when the spur gear end **830** abuts the articulation control assembly housing portion surface **832**, the spur gear **806** engages with the handle gear teeth **1100** (FIG. **11**A). In particular, the gear housing gear teeth **900** of **826** engage with the spline gear teeth **902** of **822**.

In an embodiment, a user can use the handle gear **104** in conjunction with the articulation control assembly **800** to control articulation of the distal end of the handheld surgical instrument **100** via the articulation portion **112**. In the embodiment shown with reference to FIG. **8**A, the articulation control assembly **800** can be configured to articulate the distal end of the handheld surgical instrument **100** via the articulation portion **112** such that the cutting implement **108** moves along the direction X. To further illustrate, now making reference to FIG. **11**A, a view of the handle gear **104** in cooperation with the spur gear **806** is shown in accordance with an embodiment. The handle gear **104** can include gear teeth **1100** which can engage with gear teeth **1102** of the spur gear **806** when a user rotates the handle gear **104** about a pivot **1104**. In an embodiment, when a user pivots the handle gear **104** about the pivot **1104** along the direction Y, the handle gear teeth **1100** can engage with the spur gear teeth **1102**, thereby causing the spur gear **806** to rotate along the direction X. When the spur gear **806** rotates along the direction X, the flat pull wire **208**A can move along the direction B while the flat pull wire **208**B can move along the direction A. More specifically, rotation of the spur gear **806** along the direction X can cause rotation of the articulator **826** along the direction X, thereby causing movement of the flat pull wire **208**A along the direction B and movement of the flat pull wire **208**B along the direction A. By virtue of the flat pull wire **208**A moving along the direction B and the flat pull wire **208**B moving along the direction A, the distal end of the handheld surgical instrument **100** articulates via the articulation portion **112** such that the cutting implement **108** moves along the direction X. For example, the distal end of the handheld surgical instrument **100** can articulate via the articulation portion **112** such that the cutting implement **108** can move from the substantially flat position shown with reference to FIG. **1** to an articulated position along the direction X, as shown with reference to FIGS. **4**A and **4**B and therefore have positive articulation. It should be noted that positive articulation is not limited to what is shown with reference to FIGS. **4**A and **4**B for purposes of this disclosure. In particular, the distal end of the handheld surgical instrument **100** can articulate via the articulation portion **112** such that there is more or less articulation than is shown with reference to FIGS. **4**A and **4**B during positive articulation.

In addition to articulating the distal end of the handheld surgical instrument **100** via the articulation portion **112** such that the cutting implement **108** moves along the direction X, the articulation control assembly **800** can be used for articulation along the direction Y. In particular, a user can move the actuating mechanism along a direction B such that an end **834** of the spur gear **806** contacts a surface **836** of the articulation control assembly housing, as shown with reference to FIG. **8**B. Here, when the spur gear end **834** contacts the articulation control assembly housing surface **836**, the spline **824** can engage with the handle gear spline **812**, as shown with reference to FIG. **8**B.

In the embodiment shown with reference to FIG. **8**B, the articulation control assembly is configured to articulate the distal end of the handheld surgical instrument **100** via the articulation portion **112** such that the cutting implement **108** moves along the direction Y. To further illustrate, now making reference to FIG. **11**B, a view of the handle gear **104** in cooperation with the spur gear **808** is shown in accordance with an embodiment. The handle gear **104** can include gear teeth **1106** which can engage with gear teeth **1108** of the spur gear **808** when a user rotates the handle gear **104** about a pivot **1104**. In an embodiment, when a user pivots the handle gear **104** about the pivot **1104** along the direction X, the handle gear teeth **1106** can engage with the spur gear teeth **1108**, thereby causing the spur gear **808** to rotate along the direction Y. When the spur gear **808** rotates along the direction Y, the flat pull wire **208**A moves along the direction A while the flat pull wire **208**B moves along the direction B. By virtue of the flat pull wire **208**A moving along the direction A and the flat pull wire **208**B moving along the direction B, the distal end of the handheld surgical instrument **100** articulates via the articulation portion **112** such that the cutting implement **108** moves along the direction Y. For example, the distal end of the handheld surgical instrument **100** can articulate via the articulation portion **112** such that the cutting implement **108** can move from the substantially curved position shown with reference to FIGS. **4**A and **4**B towards a position that is less curved than either of the positions shown with reference to FIGS. **4**A and **4**B, such as a negative articulation. Moreover, in accordance with the embodiments shown with regards to FIGS. **8**B and **11**B, the cutting implement **108** can be moved from the position shown with reference to FIG. **4**A to the position shown with reference to FIG. **4**B. It should be noted that articulation is not limited to what is shown with reference to FIGS. **4**A and **4**B. In particular, the distal end of the handheld surgical instrument **100** can articulate via the articulation portion **112** such that there is more or less articulation that is shown with reference to FIGS. **4**A and **4**B.

In addition to the articulation control assembly **800** shown with reference to FIGS. **8**A and **8**B, alternative articulation control assemblies can be used to articulate the distal end of the handheld surgical instrument **100** via the articulation portion **112** such that the cutting implement **108** moves along the directions X and Y. For example, the handheld surgical instrument **100** can include a ratchet articulation control assembly **1200** to articulate the distal end of the handheld surgical instrument **100** via the articulation portion **112**, as shown with reference to FIG. **12**.

In an embodiment, the ratchet control assembly **1200** can include the nose cone **110** along with an articulator **1202**, a cam **1204**, and a switch lever **1206**. It should be noted that only an upper half of the nose cone **110** is shown in FIG. **12**. In addition, the ratchet control assembly **1200** can include an outer flex gear **1208**, a handpiece idler gear **1210**, and a nose cone idler gear **1212** where the nose cone idler gear **1212** can be coupled to the nose cone **110** such that rotation of the nose cone **110** can rotate the nose cone gear **1212**. Specifically, the nose cone **110** can include gear teeth **1214** and the nose cone idler gear **1212** can include gear teeth **1216** that complement and mesh with the nose cone gear teeth **1214**. In an embodiment, the articulator **1202** can be a ratchet gear. Furthermore, the nose cone idler gear **1212** can couple with the articulator **1202** via the nose cone idler gear teeth **1216** and ratchet gear **1218** of the articulator **1202**. In an embodiment, the cam **1204**, the switch lever **1206**, and the handpiece idler gear **1210** can be operatively coupled with each other via ratchet control assembly shafts **1220** and **1221**, as shown with reference to FIG. **12**.

In the embodiment shown with reference to FIG. **12**, the handpiece idler gear **1210** can engage with teeth **1222** of the outer flex gear **1208** and the ratchet gear **1218** via gear teeth **1224**. Moreover, the outer flex gear **1208** can couple with the cutting implement **108** via the shaft **107**. Thus, when a user rotates the nose cone **110**, the nose cone idler gear **1212** can also rotate via the nose cone gear teeth **1214** and the nose cone idler gear teeth **1216**. As the nose cone idler gear **1212** rotates, by virtue of the nose cone idler gear **1212** being operatively coupled with the handpiece idler gear **1210** via the ratchet gear **1218** and the handpiece idler gear teeth **1224**, the handpiece idler gear **1210** can also rotate. Furthermore, as the handpiece idler gear **1210** rotates, by virtue of the handpiece idler gear **1210** being operatively coupled with the outer flex gear **1208** via the outer flex gear teeth **1222** and the handpiece idler gear teeth **1224**, the outer flex gear **1208** can also rotate (FIG. **16**A). Moreover, in the configuration shown with reference to FIG. **12**, as the outer flex gear **1208** rotates, the actuator **1202** can also rotate.

Since the outer flex gear **1208** couples with the cutting implement **108**, as the outer flex gear **1208** rotates, the cutting implement **108** also rotates. Therefore, when the cutting implement **108** rotates, the articulator **1202** also rotates. As noted above, the ratchet control assembly **1200** can be used to articulate the distal end of the handheld surgical instrument **100** via the articulation portion **112**. However, when the articulator **1202** rotates with the cutting implement **108**, articulation of the distal end of the handheld surgical instrument **100** via the articulation portion **112** may not occur.

In order to facilitate articulation, the ratchet control assembly **1200** operatively cooperates with the flat pull wires **208**. In particular, as shown with reference to FIGS. **12** and **13**, the flat pull wires **208** can feed through the articulator **1202** and anchor with the articulator **1202** at the proximal end of the handheld device **100** via pull wire anchors **1226**A and **1226**B. In an embodiment, when the articulator **1202** rotates along the direction X, the distal end of the handheld surgical instrument **100** can articulate along the direction X such that the articulation portion **112** can have positive articulation as described herein.

In an embodiment, in order to allow rotation of the articulator **1202** along the direction X, the ratchet control assembly **1200** can include pawls **1400** and **1402** that can engage with gear teeth **1404** of the articulator **1202**, as shown with reference to FIG. **14**. In the embodiment shown with reference to FIG. **14**, when the articulator **1202** rotates in the direction X, in order to allow articulation of the distal end of the handheld device **100**, the pawl **1400** engages with the ratchet gear teeth **1404** while the pawl **1402** disengages with the ratchet gear teeth **1404**. In an embodiment, the pawl **1400** can be caused to engage with the ratchet gear teeth **1404** by moving the cam **1204** via the switch lever **1206** into the configuration shown with reference to FIG. **14**. In an embodiment, when the cam **1204** is moved into the position shown with reference to FIG. **14**, the handpiece idler gear **1210** disengages from the outer flex gear **1208** such that when the nose cone **10** is rotated, the outer flex gear **1208** may not rotate, such that the cutting implement **108** also does not rotate. In order to rotate the articulator **1202** along the direction X, a user can rotate the nose cone **110** along the direction X. As the user rotates the nose cone **110** along the direction X, the nose cone idler gear **1212** can be rotated via the nose cone gear teeth **1214** and the nose cone idler gear teeth **1216**. As the nose cone idler gear teeth **1216** rotate, the nose cone idler gear teeth **1216** can rotate the ratchet gear teeth **1218**, thereby rotating the articulator **1202** along the direction X. In particular, by virtue of the pawl **1402** being spaced away from the articulator **1202** and not in contact with the articulator **1202**, the articulator **1202** can rotate in the direction X. Moreover, when the pawl **1400** is in the configuration shown with reference to FIG. **14**, since the outer flex gear **1208** is disengaged from the handpiece idler gear **1210**, the outer flex gear **1208** along with the cutting implement **108** may not rotate. In an embodiment, since only the articulator **1202** and the flat pull wire **208** rotate, the flat pull wire **208**A can be pulled along the direction A while the flat pull wire **208**B can be pushed along the direction B. As the flat pull wire **208**A is pulled along the direction A and the pull wire B is pushed along the direction B, the distal end of the handheld surgical instrument **100** can articulate via the articulation portion **112** such that the articulation portion **112** can have positive articulation as described herein.

Returning attention to FIGS. **12** and **13**, when the articulator **1202** rotates along the direction Y, the distal end of the handheld surgical instrument **100** can articulate along the direction Y such that the cutting implement **108** can have negative articulation as described herein. In an embodiment, in order to allow rotation of the articulator **1202** along the direction Y, the pawl **1402** can engage with ratchet gear teeth **1404**, as shown with reference to FIG. **15**. In the embodiment shown with reference to FIG. **15**, when the articulator **1202** rotates in the direction Y, in order to articulate the distal end of the handheld device **100**, the pawl **1402** engages with the ratchet gear teeth **1404** while the pawl **1400** disengages with the ratchet gear teeth **1404**. In an embodiment, by virtue of the pawl **1400** being spaced away from the articulator **1202** and not in contact with the articulator **1202**, the articulator **1202** may rotate in the direction Y. In an embodiment, the pawl **1402** can be caused to engage with the ratchet gear teeth **1404** by moving the cam **1204** via the switch lever **1206** into the configuration shown with reference to FIG. **15**. In an embodiment, when the cam **1204** is moved into the position shown with reference to FIG. **15**, the handpiece idler gear **1210** disengages from the outer flex gear **1208** such that when the nose cone **10** is rotated, the outer flex gear **1208** may not rotate, such that the cutting implement **108** also does not rotate. In order to rotate the articulator **1202** along the direction Y, a user can rotate the nose cone **110** along the direction Y. As the user rotates the nose cone **110** along the direction Y, the nose cone idler gear **1212** can be rotated via the nose cone gear teeth **1214** and the nose cone idler gear teeth **1216**. As the nose cone idler gear teeth **1216** rotate, the nose cone idler gear teeth **1216** rotate the ratchet gear teeth **1218**, thereby rotating the articulator **1202** along the direction Y. By virtue of the pawl **1400** being spaced away from the articulator **1202** and not in contact with the articulator **1202**, the articulator **1202** may rotate in the direction Y. Moreover, when the pawl **1402** is in the configuration shown with reference to FIG. **15**, since the outer flex gear **1208** is disengaged from the handpiece idler gear **1210**, the outer flex gear **1208** along with the cutting implement **108** may not rotate. In an embodiment, since only the articulator **1202** and the flat pull wire **208** rotate, the flat pull wire **208**A is pushed along the direction B while the flat pull wire **208**B is pulled along the direction A. As the flat pull wire **208**A is pushed along the direction B and the pull wire B is pulled along the direction A, the articulation portion **112** can move such that the articulation portion **112** can have negative articulation as described herein.

Moreover, in an embodiment, both of the pawls **1400** and **1402** are disengaged with the articulator **1202** as shown with reference to FIGS. **16**A and **16**B. In this embodiment, when each of the pawls **1400** and **1402** are disengaged, the cam **1204** has the position shown with reference to FIG. **16**B, such that the handpiece idler gear **1210** engages with the outer flex gear **1208** such that the articulator **1202** rotates with the outer flex gear **1208**. When the articulator **1202** rotates with the outer flex gear **1208** and the cutting implement **108**, articulation of the articulation portion **112** does not occur since the pull wires **208**A and **208**B do not move in the directions A and B.

As described above, the articulation portion **112** can include slits **200** and **202** configured to facilitate articulation of the cutting implement **108**. In accordance with a further embodiment, the articulation portion **112** can include alternative configurations that can allow for articulation of the cutting implement **108**. For example, FIG. **17** shows an embodiment where the articulation portion **112** can include compressible springs **1700** where the flat pull wires **208** are disposed within coils of the compressible springs **1700**. In an embodiment, the hypotube **1700** may be formed from any rigid material, such as metal, plastic, or the like. In this embodiment, the handheld surgical instrument **100** includes a hypotube section **1702** having passageways **1704** disposed therein through which the flat pull wires **208** may pass from a proximal end of the handheld surgical instrument **100** to the distal end of the handheld surgical instrument **100**. In an embodiment, the articulation portion **1700** may include pull wire anchor points **1706**. In an embodiment, the flat pull wires **208** may anchor with the cutting implement **108** at the anchor points **1700** with soldering, spot welding, threaded fasteners, rivets, or the like. Furthermore, the flat pull wires **208** can be anchored to the cutting implement **108** at the anchoring area with an adhesive, such as an epoxy, a glue, or the like. The embodiments shown with reference to FIG. **17** can be used with the articulation control assembly **800** or the ratchet control assembly **1200** such that either of the assemblies **800** or **1200** may be used to articulate the cutting implement **108** using the articulation portion **112** described with reference to FIG. **17**. Moreover, in an embodiment, a heat shrink outer tube **1708** may be disposed about an outer periphery of the hypotube section **1702**. While the heat shrink outer tube **1708** is shown covering a portion of the hypotube section **1702**, the heat shrink outer tube **1708** can be configured to completely cover the hypotube section **1702**.

In addition to the embodiment shown with regards to FIG. **17**, the articulation portion **112** can also have the configuration shown with reference to FIGS. **18-20**. In the embodiment shown with regards to FIG. **18**, the articulation portion **112** can include a tube **1800** that can include slots **1900** and **1902** (FIG. **19**), where the slots **1900** and **1902** are configured to allow for positive and negative articulation of the cutting implement **108** as discussed herein and with reference to FIGS. **4**A and **4**B. As may be seen with reference to FIG. **19**, the slots **1900** are disposed on a first side of the tube **1800** while the slots **1902** are disposed on a second side of the tube opposite the slots **1900**. Moreover, as may be seen with reference to FIG. **19**, the slots **1900** are offset from the slots **1902**.

In an embodiment, the flat pull wires **208** may be disposed at an inner portion of the tube **1800** and couple with anchor rings **1802**. The anchor rings **1802** can be standoffs disposed at the cutting implement **108**. In an embodiment, the anchor rings **1802** can include an annular stand-off, similar to a bearing, to support and center the rotating blade **1804**. Moreover, the anchor rings **1802** serve to anchor the pull wires disclosed herein and also to support or center or support and center the cutting blades disclosed herein. Furthermore, the anchor ring **1802** can be fully annular or interrupted. In an embodiment, the anchor rings **1802** can interface with the cutting implement **108** with soldering, spot welding, threaded fasteners, rivets, or the like. Furthermore, the anchor rings **1802** can interface with the cutting implement **108** at the anchoring area with an adhesive, such as an epoxy, a glue, or the like. Likewise, the flat pull wires **208** can anchor with the cutting implement **108** at the anchor rings **1802** with soldering, spot welding, threaded fasteners, rivets, or the like. Furthermore, the flat pull wires **208** can be anchored to the cutting implement **108** at the anchor rings **1802** with an adhesive, such as an epoxy, a glue, or the like. In addition, a rotating blade **1804** can be disposed within the tube **1800**. It should be noted that all of the embodiments disclosed herein can include the rotating blade **1804** in the configuration with reference to FIG. **18**. In an embodiment when the handheld surgical instrument **100** is being used for tissue resection, the rotating blade **1804** can rotate within the tube **1800**. In an embodiment, the rotating blade **1804** can include an inner blade edge (not shown), which defines an inner blade window (not shown). Moreover, the inner blade edge, in conjunction with the cutting implement **108**, can be used for resection during use of the handheld surgical instrument **100**. In the embodiment shown with reference to FIGS. **18-20**, the tube **1800** can be covered with a heat shrink outer tube **2000** that couples with a joint **2002**. In an embodiment, the joint **2002** can function to center the heat shrink outer tube **2000** over the articulation portion **112**. The embodiments shown with reference to FIGS. **18-20** can be used with the articulation control assembly **800** or the ratchet control assembly **1200** such that either of the assemblies **800** or **1200** can be used to articulate the cutting implement **108** using the articulation portion **112** described with reference to FIGS. **18-20**.

In addition to the embodiments shown with reference to FIGS. **17-20**, the articulation portion **112** may have the configuration shown with reference to FIGS. **21** and **22**. In this embodiment, the articulation portion **112** can include an articulation member **2100**, which can be a compression spring or a semi-rigid ductile polymer, such as plastic. In this embodiment, the flat pull wire **208** is disposed at an outer surface of the shaft **107** and extends from the proximal portion of the handheld surgical instrument **100** to the cutting implement **108**. The embodiments shown with reference to FIGS. **21** and **22** can be used with the articulation control assembly **800** or the ratchet control assembly **1200** such that either of the assemblies **800** or **1200** can be used to articulate the cutting implement **108** using the articulation portion **112** described with reference to FIGS. **21** and **22**. Thus, in an embodiment, either of the assemblies **800** and **1200** can be used to control the flat pull wire **208** such that the articulation member **2100** articulates as shown with reference to FIG. **21**. More specifically, the flat pull wire **208** may be pulled along the direction B, thereby causing the cutting implement **108** to articulate as shown with reference to FIG. **22**. In some embodiments, the pull wire **208** can be inside the tube **107** or between the tube **107** and another tube disposed about the tube **107**.

In addition to the embodiments discussed above, the cutting implement **108** can be articulated in accordance with embodiments disclosed with references to FIGS. **23-33**. Now making reference to FIG. **23**, an alternative embodiment of the articulation portion **112** is shown. In this embodiment, the articulation portion **112** can be disposed on a shaft **2300** and can include slits **2302**. In an embodiment, the slits **2302** can allow for flexing of the articulation portion **112** thereby allowing positive and negative articulation of the cutting implement **108**, as discussed herein and shown with reference to FIGS. **4**A and **4**B. In particular, the shaft **2300**, along with the articulation portion **112**, can include passageways **2400** that are configured to receive the flat pull wires **208**, as shown with regards to FIGS. **24** and **25**. Therefore, using the flat pull wires **208**, the cutting implement **108** can be articulated such that the cutting implement **108** can have positive articulation as described herein. Moreover, using the flat pull wires **208**, the cutting implement **108** can be articulated from a curved position such that the cutting implement **108** can have negative articulation as described herein. In an embodiment, the flat pull wires **208** can be controlled with the articulation control assembly **800** or the ratchet control assembly **1200** discussed above.

In an embodiment, the slits **2302** can be configured to allow articulation of the cutting implement **108** as discussed above. In an embodiment, the slits **2302** can have a width W_{2500} that can be in a range of about 0.0038 inches to about 0.0046 inches. In an embodiment, the slits **2302** can have a width W_{2500} that can be in a range of about 0.005 inches to about 0.150 inches. In an embodiment, by virtue of the slits **2302** having the width W_{2500}, the slits **2302** can allow articulation of the articulation portion **112**.

Now making reference to FIGS. **26** and **27**, further methods of articulating the cutting implement **108** are shown in accordance with an alternative embodiment of the present disclosure. In accordance with an embodiment, the handheld surgical instrument **100** can include a shaft **2700** that can include an articulation portion **2702**, as shown with reference to FIGS. **26** and **27**. In an embodiment, each of the shaft **2700** and the articulation portion **2702** can be formed of a semi-rigid material that is bendable. Examples of a material that can be used for the each of the shaft **2700** and the articulation portion **2702** can include stainless spring steel, plastic, a shape-memory material, such as nitinol or super-elastic nitinol, or any other type of polymer having bendable properties. In an embodiment, the cutting implement **108** can be disposed at a distal end of the shaft **2700** such that when the handheld surgical device **100** includes the shaft **2700** and the articulation portion **2702**, the handheld surgical device **100** can be used for tissue resection, as discussed above.

In the embodiment shown with regards to FIGS. **26** and **27**, an articulating shaft **2704** may be inserted into a passageway **2800** (FIGS. **28** and **29**). In an embodiment, the articulating shaft **2704** may be formed of a rigid a material, including a metal alloy such as steel or aluminum, a rigid polymer, or the like, and may traverse portions of both the shaft **2700** and the articulation portion **2702**. In an embodiment, due to the rigidity of the articulating shaft **2704**, when the articulating shaft **2704** completely traverses the articulation portion **2702** via the passageway **2800** in a first position, the shaft **2700** and the articulation portion **2702** have a substantially straight configuration, as shown with reference to FIG. **26**. Here, a natural position of the articulation portion **2702** may be in a bent configuration as shown with reference to FIG. **27**. Thus, when the articulating shaft **2704** is removed from the articulation portion **2702** in a second position, the articulation portion **2702** can have the bent configuration as shown with reference to FIG. **27**. When the articulating shaft **2700** traverses the articulation portion **2702**, either completely or partially, the articulating shaft **2700** can cause the articulation portion **2702** to have a straight configuration as shown with regards to FIG. **26**. Moreover, the articulating shaft **2700** can be moved in the direction A or the direction B such that the cutting implement **108** can move from the substantially curved position shown with reference to FIGS. **4**A and **4**B towards a position that is less curved than either of the positions shown with reference to FIGS. **4**A and **4**B. It should be noted that articulation is not limited to what is shown with reference to FIGS. **4**A and **4**B. In particular, the distal end of the handheld surgical instrument **100** can articulate via the articulation portion **112** such that there is more or less articulation that is shown with reference to FIGS. **4**A and **4**B. In addition, using the articulating shaft **2700**, the cutting implement **108** can be moved from a substantially flat position to one of the positions shown with reference to FIGS. **4**A and **4**B where the articulating shaft can be moved along the direction A.

In an embodiment, the articulating shaft **2704** can be connected to a control assembly, such as the articulation control assembly **800** or the ratchet control assembly **1200** discussed above. In an embodiment, a proximal end of the articulating shaft **2704** can be coupled to the flat pull wire **208** and can be controlled via the articulation control assembly **800**. In this embodiment, only a single flat pull wire **208**, such as the flat pull wire **208**A, would be used with the articulation control assembly **800** since articulation of the shaft **2700** and the articulation portion **2702** are controlled with a single implement, the articulation shaft **2704**. For example, the articulation control assembly **800** would only include the flat pull wire **208**A where the articulation control assembly **800** only controls the flat pull wire **208**A as discussed above. Moreover, in this embodiment, only a single flat pull wire **208**, such as the flat pull wire **208**A, would be used with the ratchet control assembly **1200** since articulation of the shaft **2700** and the articulation portion **2702** are controlled with a single implement, the articulation shaft **2404**. For example, the articulation control assembly **1200** would only include the flat pull wire **208**A where the articulation control assembly **1200** only controls the flat pull wire **208**A as discussed above. A rack-and-pinion system may also be used to control movement of the articulating shaft **2704**, in accordance with an embodiment.

Now making reference to FIGS. **30** and **31**, further methods of articulating the cutting implement **108** are shown in accordance with an alternative embodiment of the present disclosure. The handheld surgical instrument **100** can include the shaft **2700** and the articulating portion **2702** along with a straight tube **3000**. The straight tube **3000** can be formed of any rigid material. Examples can include any type of metal alloy, such as stainless steel or aluminum or the like, or any rigid polymer. In an embodiment, due to the rigidity of the straight tube **3000**, when the straight tube **3000** completely traverses or partially traverses the articulation portion **2702**, the shaft **2700** and the articulation portion **2702** have a substantially straight configuration, as shown with reference to FIG. **30**. Here, a natural position of the articulation portion **2702** may be in a bent configuration as shown with reference to FIG. **31** and discussed above. When the straight tube **3000** traverses the articulation portion **2702**, either completely or partially, the straight tube

3000 can cause the articulation portion 2702 to have a straight configuration in a first position as shown with regards to FIG. 30.

In an embodiment, the straight tube 3000 can slidingly engage with the shaft 2700 and the articulation portion 2702 such that a user can slide the straight tube 3000 along the directions A and B. In an embodiment, when a user slides the straight tube 3000 along the direction B from the configuration shown with respect to FIG. 30, the cutting implement 108 may articulate into the configuration in a second position shown with reference to FIG. 31. Furthermore, the straight tube 3000 can be moved in the direction A or the direction B such that the cutting implement 108 can move from the substantially curved position shown with reference to FIGS. 4A and 4B towards a position that is less curved than either of the positions shown with reference to FIGS. 4A and 4B. It should be noted that articulation is not limited to what is shown with reference to FIGS. 4A and 4B. In particular, the distal end of the handheld surgical instrument 100 can articulate via the articulation portion 112 such that there is more or less articulation that is shown with reference to FIGS. 4A and 4B. In an embodiment, the straight tube 3000 can include a knob that can allow for movement of the straight tube 3000 along the directions A and B. In addition, using the straight tube 3000, the cutting implement 108 can be moved from a substantially flat position to one of the positions shown with reference to FIGS. 4A and 4B, where the straight tube 3000 can be moved along the direction A.

Additionally, the straight tube 3000 can threadingly engage with the articulation portion 2702. In particular, the articulation portion 2702 can include threads 3200 (FIG. 32) disposed about an outer surface of the articulation portion 2702. In this embodiment, the straight tube 3000 can include threads 3300 (FIG. 33) disposed at an inner surface thereof that complement the articulation portion threads 3200. In this configuration, a user can rotate the straight tube 3000 along the direction A or the direction Y. In an embodiment, when a user twists the straight tube 3000 along the direction X, the straight tube moves along the direction A. In an embodiment, when a user twists the straight tube 3000 along the direction Y, the straight tube moves along the direction B. Therefore, a user may control articulation of the cutting implement 108 by twisting the straight tube 3000 in either the direction X or the direction Y. In an embodiment, the straight tube 3000 can include a knob that can allow for twisting the straight tube along the directions X and Y.

As discussed above, during articulation of the cutting implement 108, the flat pull wires 208 can be controlled with the articulation control assembly 800 or the ratchet control assembly 1200. The flat pull wires 208 may also be controlled with other assemblies, as shown with reference to FIGS. 34-40. Making reference to FIG. 34, an articulation control assembly 3400 is shown in accordance with an embodiment of the present disclosure. In an embodiment, the articulation control assembly 3400 can include an articulator 3402 disposed about a pivot 3404 and can be disposed within the handpiece 102. The articulator 3402 can couple with an actuator mechanism 3406 via a biasing mechanism neck 3408, as shown with reference to FIG. 34. The actuating mechanism 3406 can include an actuator 3410 operatively coupled with a locking mechanism 3412 via an actuator neck 3414.

In an embodiment, the flat pull wires 208A and 208B couple with the articulator 3402 at anchor points 3416. In an embodiment, the flat pull wires 208A and 208B may anchor with the anchor points 3416 at an anchoring area 3418 with soldering, spot welding, threaded fasteners, rivets, or the like. In an embodiment, the articulator 3402 is configured to rotate about the biasing mechanism pivot 3404 along the directions X and Y. In an embodiment, when the articulator 3402 rotates about the biasing mechanism pivot 3404 along the direction Y, the articulator 3402 rotates the flat pull wires 208A and 208B along the direction Y. More specifically, as the articulator 3402 rotates along the direction Y, the pull wire 208A moves along the direction B and the pull wire 208B moves along the direction A.

In an embodiment, when the articulator 3402 rotates along the direction Y, the articulator 3402 can rotate the flat pull wires 208A and 208B such that the cutting implement 108 can have positive articulation as described herein. The articulator 3402 can also rotate along the direction X about the biasing mechanism pivot 3404. When the articulator 3402 rotates along the direction X, the flat pull wires 208A and 208B can also rotate along the direction X such that the cutting implement 108 can have negative articulation as described herein.

In an embodiment, the articulator 3402 can be caused to rotate about the biasing mechanism pivot 3404 with the actuator mechanism 3406. The locking mechanism 3412 can include teeth 3420 that are configured to engage with teeth 3422 of the articulation control assembly 3400. In an embodiment, when the locking mechanism teeth 3420 are engaged with the articulation control assembly teeth 3422 as shown with reference to FIG. 34, the articulation control assembly 3400 can be in a locked position. In order to cause rotation of the articulator 3402 in either direction X or the direction Y, a user can move the actuator 3410 along a direction C such that the locking mechanism teeth 3420 are disengage with the articulation control assembly teeth 3422. Upon disengagement, the user may rotate the articulator 3402 about the biasing mechanism pivot 3404 along the direction X or the direction Y, thereby controlling articulation of the cutting implement 108.

Besides the embodiment shown with reference to FIG. 34, articulation of the cutting implement 108 can be controlled via an articulation control assembly 3500 disposed within the handheld surgical instrument housing 114, as shown with respect to FIGS. 35 and 36A. In an embodiment, the articulation control assembly 3500 can include actuator slides 3502 and 3600 that are configured to slide along guides, such as a guide 3504, within the handheld surgical instrument housing 114. Each of the actuator slides 3502 and 3600 can couple with the flat pull wires 208 via anchoring points 3602 as shown with reference to FIG. 36A. In an embodiment, the flat pull wires 208 can anchor with the anchoring points 3602 at anchoring areas 3604 with soldering, spot welding, threaded fasteners, rivets, or the like.

In an embodiment, movement of the actuator slides 3502 and 3600 can cause articulation of the cutting implement 108 via the flat pull wires 208. For example, movement of the actuator slide 3502 along the direction A can cause articulation of the cutting implement 108 such that the cutting implement 108 can have positive articulation as described herein. In this example, when the movement of the actuator slide 3502 along the direction A causes positive articulation of the cutting implement 108, movement of the actuator slide 3502 along the direction B can cause articulation of the cutting implement 108 such that the cutting implement 108 can have negative articulation as described herein. Moreover, when the movement of the actuator slide 3502 along the direction A causes positive articulation of the cutting implement 108, movement of the actuator slide 3600 along the direction B can cause positive articulation of the cutting implement 108.

In addition, movement of the actuator slide **3502** along the direction A can cause articulation of the cutting implement **108** such that the cutting implement **108** can have negative articulation as described herein. In embodiments where movement of the actuator slider **3502** along the direction A causes negative articulation of the cutting implement, movement of the actuator slider **3600** along the direction B can cause articulation of the cutting implement **108** such that the cutting implement **108** can have negative articulation as described herein.

Moreover, the articulation control assembly **3500** can be secured in various positions as shown with reference to FIGS. **36**B and **36**C. The actuator slide **3502** can include pins **3606** and **3608** that can engage with notches **3612** of the actuator slide **3502**, as shown with reference to FIGS. **36**B and **36**C. The articulation control assembly **3500** can include a spring **3610**, such as a leaf spring, that can bias a pin **3614** that can be coupled to the articulation control assembly **3500** along a direction Z_1. In an embodiment, the pins **3606** and **3608** can be configured to rest within notches **3612** when the spring **3610** biases the articulation control assembly **3500** along the direction Z_1. When the pins **3606** and **3608** are disposed within the notches **3612**, the articulation control assembly **3500** is locked into position. In order to articulate the articulation portion **112**, the articulation control assembly **3500** can be moved in a direction Z_2, thereby moving the pins **3606** and **3608** out of the notches **3612**. Moreover, the articulation control assembly can be moved along either of the directions A and B while the articulation control assembly **3500** can be moved in a direction Z_2. When the articulation control assembly **3500** is no longer moved in the direction Z_2, the spring **3610** can bias the articulation control assembly **3500** along the direction Z_1, whereby the pins **3606** and **3608** are moved into other notches **3612**.

In addition to the embodiment shown with reference to FIGS. **35** and **36**A, articulation of the cutting implement **108** can be controlled via an articulation control assembly **3700** disposed within the handheld surgical instrument housing **114**, as shown with respect to FIGS. **37**A and **37**B. In an embodiment, the articulation control assembly **3700** can include an articulator **3702** disposed about a pivot **3704** along with biasing mechanism interfaces **3706** disposed about a periphery of the articulator **3702**. In an embodiment, the flat pull wires **208** can couple with the articulator **3702** at an anchor point **3708**. In an embodiment, the flat pull wires **208** may anchor with the anchor point **3708** with soldering, spot welding, threaded fasteners, rivets, or the like.

The biasing mechanism can be used to positively and negatively articulate the cutting implement **108**. In particular, a user may rotate the articulator **3702** along either the direction X or the direction Y in order to move the pull wire wires **208**A and **208**B along the directions A and B. In an embodiment, the articulator **3702** can be a rotatable thumb wheel that rotates along the directions X and Y. In an embodiment, the biasing mechanism interfaces **3706** can protrude from the housing **114** at areas **3710**. Thus, a user can engage the biasing mechanism interfaces **3706** at the areas **3710** and rotate the articulator **3702** along the directions X and Y. In particular, the articulation control assembly **3700** can include planetary gears **3712** and **3714** which can cooperate with each other and teeth **3716** to effectuate articulation of the articulation portion **112**.

To further illustrate, in an example, rotation of the articulator **3702** along the direction X can cause movement of the flat pull wire **208**A along the direction A and can cause positive articulation of the cutting implement **108** as described herein. In this example, a user may engage the biasing mechanism interfaces **3706** and rotate the articulator **3702** along the direction X. As the articulator **3702** rotates along the direction X, the flat pull wire **208**A moves along the direction A while the flat pull wire **208**B moves along the direction B via movement of the planetary gears **3712** and **3714**. In this example, as the flat pull wire **208**A moves along the direction A, the cutting implement **108** can articulate such that the cutting implement **108** can have positive articulation as described herein. Moreover, in this example, as a user rotates the articulator **3702** along the direction Y, the flat pull wire **208**A moves along the direction B while the flat pull wire **208** moves along the direction A via movement of the planetary gears **3712** and **3714**. Here, as the flat pull wire **208**A moves along the direction B, the cutting implement **108** can articulate such that the cutting implement **108** can have negative articulation as described herein.

In a further example, movement of the flat pull wire **208**A along the direction A can cause negative articulation of the cutting implement **108** as described herein. In this example, a user can engage the biasing mechanism interfaces **3706** and rotate the articulator **3702** along the direction X. As the articulator **3702** rotates along the direction X, the flat pull wire **208**A moves along the direction A while the flat pull wire **208**B moves along the direction B via movement of the planetary gears **3712** and **3714**. In this example, as the flat pull wire **208**A moves along the direction A, the cutting implement **108** can articulate such that the cutting implement **108** can have negative articulation as described herein. Moreover, in this example, as a user rotates the articulator **3702** along the direction Y, the flat pull wire **208**A moves along the direction B while the flat pull wire **208** moves along the direction A. Here, as the flat pull wire **208**A moves along the direction B, the cutting implement **108** can articulate such that the cutting implement **108** can have positive articulation as described herein.

In addition to the embodiment shown with reference to FIGS. **37**A and **37**B, articulation of the cutting implement **108** can be controlled via an articulation control assembly **3800** disposed within the handheld surgical instrument housing **114**, as shown with respect to FIGS. **38**A and **38**B. In an embodiment, the articulation control assembly **3800** can include a first biasing mechanism **3802** disposed about a pivot **3804** along with a second biasing mechanism **3806** disposed about a pivot **3808**. The first biasing mechanism **3802** can include gear teeth **3810** and the second biasing mechanism **3810** can include gear teeth **3812**, where the first biasing mechanism gear teeth **3810** can operatively engage with the second biasing mechanism gear teeth **3812**.

The first biasing mechanism **3802** and the second biasing mechanism **3806** each include anchor points **3814** and **3816** that are configured to anchor the flat pull wires **208**A and **208**B with each of the first biasing mechanism **3802** and the second biasing mechanism **3806**, respectively. In an embodiment, the flat pull wires **208**A and B may anchor with the anchor points **3814** and **3816** with soldering, spot welding, threaded fasteners, rivets, or the like. Thus, when the first biasing mechanism **3802** and the second biasing mechanism **3806** rotate, the flat pull wires **208**A and **208**B rotate with the first biasing mechanism **3802** and the second biasing mechanism **3806**.

In an embodiment, the articulation control assembly **3800** can include an actuator **3818**, which can be engaged by a user to rotate the first biasing mechanism along the direction X and the direction Y. In an embodiment, the actuator **3818** can protrude from an opening **3820** at an area **3822** of the handheld surgical instrument housing **114**, as shown with regards to FIGS. **38**A and **38**B. Here, when a user rotates the first biasing mechanism **3802** in the direction X, by virtue of the first biasing mechanism gear teeth **3810** engaging with the second biasing mechanism gear teeth **3812**, the second biasing mechanism can rotate in the direction Y. Moreover, as the first biasing mechanism **3802** rotates along direction X, by virtue of the flat pull wire **208**A being anchored to the first biasing mechanism **3802** via the anchor points **3814** and **3816**, the flat pull wire **208**A can move in the direction A. As the second biasing mechanism **3806** rotates along direction Y, by virtue of the flat pull wire **208**B being anchored to the second biasing mechanism **3806** via the anchor points **3814** and **3816**, the flat pull wire **208**B can move in the direction B. Due to the movement of the flat pull wire **208**A along the direction A along with movement of the flat pull wire **208**B along the direction B, the cutting implement **108** can articulate such that the cutting implement **108** can have positive articulation as described herein.

In addition, when a user rotates the first biasing mechanism **3802** in the direction Y using the actuator **3818**, by virtue of the first biasing mechanism gear teeth **3810** engaging with the second biasing mechanism gear teeth **3812**, the second biasing mechanism can rotate in the direction X. As the first biasing mechanism **3802** rotates along direction Y, by virtue of the flat pull wire **208**A being anchored to the first biasing mechanism **3802** via the anchor points **3814** and **3816**, the flat pull wire **208**A can move in the direction B. As the second biasing mechanism **3806** rotates along direction X, by virtue of the flat pull wire **208**B being anchored to the second biasing mechanism **3806** via the anchor points **3814** and **3816**, the flat pull wire **208**B can move in the direction A. Due to the movement of the flat pull wire **208**A along the direction B along with movement of the flat pull wire **208**B along the direction A, the cutting implement **108** can articulate such that the cutting implement **108** can have negative articulation as described herein.

In addition to the embodiment shown with reference to FIGS. **38**A and **38**B, articulation of the cutting implement **108** can be controlled via an articulation control assembly **3900** disposed within the handheld surgical instrument housing **114**, as shown with respect to FIG. **39**. In an embodiment, the articulation control assembly **3900** can include a first biasing mechanism **3902** and a second biasing mechanism **3904**. The first biasing mechanism **3902** can include a pivot **3906** about which the first biasing mechanism **3902** can rotate along the directions X and Y. The second biasing mechanism **3904** can include a pivot **3908** about which the second biasing mechanism **3904** can rotate along the directions X and Y. Each of the first biasing mechanism **3902** and the second biasing mechanism **3904** can include anchor points **3910** to which the flat pull wires **208**A and **208**B can anchor to each of the first biasing mechanism **3902** and the second biasing mechanism **3904**. In an embodiment, the flat pull wires **208**A and **208**B can anchor with the anchor points **3910** with soldering, spot welding, threaded fasteners, rivets, or the like. Thus, when the first biasing mechanism **3902** and the second biasing mechanism **3904** rotate, the flat pull wires **208**A and **208**B rotate with the first biasing mechanism **3902** and the second biasing mechanism **3806**, respectively. Moreover, each of the first biasing mechanism **3902** and the second biasing mechanism **3904** can include guides **3912**, which can respectively guide the flat pulls wires **208**A and **208**B during actuation of the first biasing mechanism **3902** and the second biasing mechanism **3904**.

In addition, the first biasing mechanism **3902** can include gear teeth **3914** and the second biasing mechanism **3904** can include gear teeth **3916**. In an embodiment, the first biasing mechanism gear teeth **3914** can cooperatively engage with the second biasing mechanism gear teeth **3916** such that when a user rotates the first biasing mechanism **3902** via an actuator **3918** along the direction X, the second biasing mechanism **3904** can rotate along the direction Y. As the first biasing mechanism **3902** rotates along the direction X, the flat pull wire **208**A can move along the direction A and the flat pull wire **208**B moves along the direction B, thereby causing positive articulation of the cutting implement **108**, as discussed herein.

In addition, when a user rotates the first biasing mechanism **3902** via the actuator **3918** along the direction Y, the second biasing mechanism **3904** can rotate along the direction X. As the first biasing mechanism **3902** rotates along the direction Y, the flat pull wire **208**A can move along the direction B and the flat pull wire **208**B moves along the direction A, thereby causing negative articulation of the cutting implement **108**, as discussed herein. It should be noted that the embodiments discussed with reference to FIGS. **8-10** may be implemented with the embodiments discussed with reference to FIGS. **38** and **39**.

Besides the embodiment shown with reference to FIG. **39**, articulation of the cutting implement **108** can be controlled via an articulation control assembly **4000** disposed within the handheld surgical instrument housing **114**, as shown with respect to FIG. **40**. In an embodiment, the articulation control assembly **4000** can include rocker switches **4002**A and **4002**B having a pivot **4004** about which the rocker switches **4002**A and **4002**B pivot. The rocker switches **4002**A and **4002**B can also include anchor points **4006** to which flat pull wires **208**A and **208**B anchor. The flat pull wires **208**A and **208**B can anchor to the anchor points **4006** with soldering, spot welding, threaded fasteners, rivets, or the like. A user can articulate the cutting implement **108** using the rocker switches **4002**A and **4002**B. For example, a user can rotate the rocker switch **4002**A along the direction Y while rotating the rocker switch **4002**B along the direction Y. As the user rotates the rocker switch **4002**A along the direction Y, the flat pull wire **208**A can move along the direction A. In addition, as the user rotates the rocker switch **4002**B along the direction Y, the flat pull wire **208**B can move along the direction B. As the flat pull wire **208**A moves along the direction A and the flat pull wire **208**A moves along the direction B, the cutting implement **108** can articulate such that the cutting implement **108** can have positive articulation as described herein.

As another example, a user can rotate the rocker switch **4002**A along the direction X while rotating the rocker switch **4002**B along the direction X. As the user rotates the rocker switch **4002**A along the direction X, the flat pull wire **208**A can move along the direction B. In addition, as the user rotates the rocker switch **4002**B along the direction X, the flat pull wire **208**B can move along the direction A. As the flat pull wire **208**A moves along the direction B and the flat pull wire **208**A moves along the direction A, the cutting implement **108** can articulate such that the cutting implement **108** can have negative articulation as described herein.

The above detailed description includes references to the accompanying drawings, which form a part of the detailed description. The drawings show, by way of illustration, specific examples in which the invention can be practiced. These examples are also referred to herein as "examples." Such examples can include elements in addition to those shown or described. However, the present inventor also contemplates examples in which only those elements shown or described are provided. Moreover, the present inventor also contemplates examples using any combination or permutation of those elements shown or described (or one or more aspects thereof), either with respect to a particular example (or one or more aspects thereof), or with respect to other examples (or one or more aspects thereof) shown or described herein.

In this document, the terms "a" or "an" are used, as is common in patent documents, to include one or more than one, independent of any other instances or usages of "at least one" or "one or more." In this document, the term "or" is used to refer to a nonexclusive or, such that "A or B" includes "A but not B," "B but not A," and "A and B," unless otherwise indicated. In this document, the terms "including" and "in which" are used as the plain-English equivalents of the respective terms "comprising" and "wherein." Also, in the following claims, the terms "including" and "comprising" are open-ended, that is, a system, device, article, composition, formulation, or process that includes elements in addition to those listed after such a term in a claim are still deemed to fall within the scope of that claim. Moreover, in the following claims, the terms "first," "second," and "third," etc. are used merely as labels, and are not intended to impose numerical requirements on their objects.

The above description is intended to be illustrative, and not restrictive. For example, the above-described examples (or one or more aspects thereof) can be used in combination with each other. Other examples can be used, such as by one of ordinary skill in the art upon reviewing the above description. The Abstract is provided to comply with 37 C.F.R. § 1.72(b), to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. Also, in the above Detailed Description, various features can be grouped together to streamline the disclosure. This should not be interpreted as intending that an unclaimed disclosed feature is essential to any claim. Rather, inventive subject matter can lie in less than all features of a particular disclosed example. Thus, the following claims are hereby incorporated into the Detailed Description as examples or examples, with each claim standing on its own as a separate example, and it is contemplated that such examples can be combined with each other in various combinations or permutations. The scope of the invention should be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

What is claimed is:

1. A handheld surgical instrument, comprising:

a handpiece;

a shaft extending from a proximal end of the handpiece;

a cutting implement disposed at a distal end of the shaft;

an articulation portion disposed proximate to the cutting implement, the articulation portion including:

a set of first slits; and

a set of second slits opposite the set of first slits;

a first flat pull wire disposed in the articulation portion;

a second flat pull wire disposed in the articulation portion; and

an articulation control assembly disposed at the handpiece, the articulation control assembly comprising:

an articulator that includes a first anchor point coupled with the first flat pull wire and a second anchor point coupled with the second flat pull wire, wherein the articulator moves the first flat pull wire in a first direction and moves the second flat pull wire in a second direction opposite the first direction when the articulator rotates;

a first spline;

a first actuator coupled with the first spline;

a second spline; and

a second actuator coupled with the second spline, wherein one of the first spline and the second spline engages with the articulator when the other of the first spline and the second spline disengages with the articulator.

2. The handheld surgical instrument of claim **1**, wherein the set of first slits are offset from the set of second slits.

3. The handheld surgical instrument of claim **1**, wherein the articulation portion includes a tube and the set of first slits are disposed at a first side of the tube and the set of second slits are disposed at a second side of the tube.

4. The handheld surgical instrument of claim **1**, wherein the articulator is a gear.

5. The handheld surgical instrument of claim **1**, wherein the articulator moves the first flat pull wire in the first direction and moves the second flat pull wire in the second direction opposite the first direction when the first spline engages with the articulator.

6. The handheld surgical instrument of claim **5**, wherein the articulator moves the first flat pull wire in the second direction and moves the second flat pull wire in the first direction opposite the second direction when the second spline engages with the articulator.

7. A handheld surgical instrument, comprising:

a handpiece;

a shaft extending from a proximal end of the handpiece;

a cutting implement disposed at a distal end of the shaft;

an articulation portion disposed proximate to the cutting implement, wherein the articulation portion includes compressible springs;

a first flat pull wire disposed within the articulation portion;

a second flat pull wire disposed within the articulation portion opposite the first flat pull wire; and

an articulation control assembly disposed at the handpiece, the articulation control assembly comprising:

an articulator that includes a first anchor point coupled with the first flat pull wire and a second anchor point coupled with the second flat pull wire, wherein the articulator moves the first flat pull wire in a first direction and moves the second flat pull wire in a second direction opposite the first direction when the articulator rotates;

a first spline;

a first actuator coupled with the first spline;

a second spline; and

a second actuator coupled with the second spline, wherein one of the first spline and the second spline engages with the articulator when the other of the first spline and the second spline disengages with the articulator.

8. The handheld surgical instrument of claim **7**, wherein the articulator is a gear.

9. The handheld surgical instrument of claim **7**, wherein the articulator moves the first flat pull wire in the first direction and moves the second flat pull wire in the second direction opposite the first direction when the first spline engages with the articulator.

10. The handheld surgical instrument of claim **9**, wherein the articulator moves the first flat pull wire in the second direction and moves the second flat pull wire in the first direction opposite the second direction when the second spline engages with the articulator.

11. A handheld surgical instrument, comprising:

a handpiece;

a shaft extending from a proximal end of the handpiece;
a cutting implement disposed at a distal end of the shaft;
an articulation portion disposed proximate to the cutting implement, the articulation portion including:
a set of first slits;
a set of second slits opposite the set of first slits;
a first passageway disposed in each slit of the set of first slits; and
a second passageway disposed in each slit of the set of second slits;
a first flat pull wire disposed in the first passageway and anchored adjacent to the cutting implement;
a second flat pull wire disposed in the second passageway and anchored adjacent to the cutting implement; and
an articulation control assembly disposed at the handpiece, the articulation control assembly comprising:
an articulator that includes a first anchor point coupled with the first flat pull wire and a second anchor point

coupled with the second flat pull wire, wherein the articulator moves the first flat pull wire in a first direction such that the first flat pull wire causes articulation of the cutting implement at the articulation portion and moves the second flat pull wire in a second direction opposite the first direction when the articulator rotates;

a first spline;

a first actuator coupled with the first spline;

a second spline; and

a second actuator coupled with the second spline, wherein one of the first spline and the second spline engages with the articulator when the other of the first spline and the second spline disengages with the articulator.

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